

DC-DC Converter for Electric Vehicle

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Abstract

In this work, a DC-DC converter is designed for an electric vehicle. The DC-DC converter is designed to provide 500W with a 200-400V input and a 12-15V adjustable output. Electric vehicle sales are beginning to increase in popularity and the need for DC-DC converters to siphon power from the tractive system is not yet fully satisfied, especially for single-seater class vehicles. Additionally, improving performance in efficiency without sacrificing wide input voltage range can benefit future DC-DC converter designs. In the end, a forward active clamp DC-DC converter is designed and tested. Additionally, spreadsheet calculators, LTSpice simulations, and Matlab scripts were made to assist in work for the DC-DC.

1. Introduction

Electric Vehicle (EV) sales have grown dramatically since 2010 [1]. The first mass-produced EV, the EV1, sported a range of 70 to 100 miles, while a base-model Tesla Model 3 has a range of 264 miles [2] [3] for similar costs. Improvements in total range and lifetime of EV's is mostly responsible for this massive increase in popularity. The two main ways of improving EV range is through system efficiency and improved battery technology.

System efficiency improvements are mostly due to the use and improvement of power electronics systems within the vehicle. Power electronics refers to the use of switching electronics for power conversion and distribution. There are six main categories of power converters: AC-DC converters (rectifiers), AC-AC converters, DC-AC converters, static switches, and DC-DC converters. AC-DC, DC-DC, and DC-AC converters. These play important roles for EV usage. AC-DC converters are often used for on-board charging systems, which allows the EV to be charged from the grid. DC-AC converters are typically used to drive the main motors of the vehicle, providing propulsion [4]. In most EV's, there is a high voltage system consisting of the vehicle's battery pack and drive system, and a low voltage system, containing the vehicle's media entertainment, sensors and telemetry, control unit, etc. To reduce weight and improve vehicle packaging, most EV's use a DC-DC converter with power capability on the order of kilowatts to efficiently step down the vehicle's battery pack voltage, 200V-800V, to the vehicle's low voltage system operating voltage, usually 12 or 24V [5].

There are many different types of DC-DC converters, but they can be divided into two main categories, isolated and non-isolated. Isolated designs incorporate a galvanic isolation between the input and output while non-isolated systems do not. Isolated designs offer a high

degree of safety because failures on the input side are generally isolated from the output, but non-isolated designs are typically smaller because they do not require a galvanic isolating device. A standard DC-DC converter will contain a switching device, an inductor, and input and output capacitors. In an isolated device, a transformer will typically be included as the method of galvanic isolation.

In order to increase knowledge in experience in automotive development, the Society of Automotive Engineers (SAE) at Cal Poly State University created the Formula SAE Student Design Competition (FSAE). In FSAE, students design, build and compete a single-seater race-car. Currently, there are both an Internal Combustion (IC) and EV competition. Internally, the EV race-car contains many similar components to a commercial EV, including a HV (High Voltage) to LV (Low Voltage) DC-DC converter. For race-car, development size and weight are among the most important design metrics. Reducing weight and size allows the vehicle to turn and accelerate much faster, improving the race-car's chances of winning the race. By extension, system efficiency is also very important because improving the range of a vehicle will allow designers to reduce the total volume and mass of batteries required. This project will explore the design of a DC-DC converter for an FSAE EV.

2. Background

Cal Poly's Formula SAE club, Cal Poly Racing, needs a smaller, higher voltage rated DC-DC converter because their existing solution, the RSP-500-12 [6], is only rated at 370 VDC, which is a slight 0.4 V more than the expected maximum operating voltage of 369.6 VDC. The purpose of the DC-DC converter is to use the "high-voltage" 316 V battery pack, which is used to drive the electric motor, to power all of the low voltage electronics while the car operates. During times when the high-voltage battery is disconnected, a "low-voltage" 14 V battery is used to maintain low voltage electronics. Although a low voltage battery could be used instead of a DC-DC converter, the size of such a battery is infeasible for the packaging constraints of a Formula SAE Electric vehicle.

The Formula SAE Electric vehicle can draw 500W of power continuously on the low voltage system and the high voltage battery pack voltage can vary from less than 246 V to 370 V. Additionally, the DC-DC converters must fit in a narrow enclosure next to the battery pack in order to pass high voltage isolation rules. These requirements can be summarized as a DC-DC converter that must be able to transform 200-400 VDC to 14 VDC at a power level of 500 W in the smallest package possible.

Because climate change has become a greater concern in the 21st century, the automotive industry is slowly moving towards hybrid and fully-electric vehicles. As a result, the use of DC-DC converters in electric vehicles has been a topic of development and research in the automotive industry. Most DC-DC converters used in hybrid or electric vehicles consist of both unidirectional and bidirectional DC-DC converters. The unidirectional converters generally have

been used for vehicle sensors, controls, entertainment, utility, and safety purposes. Whereas the bidirectional DC-DC converters have found their use in battery charging and regenerative braking. Many modern day DC-DC converters use soft-switching technique to achieve higher efficiency and are isolated to provide safety to the load. However, the use of transformers leads to leakage inductance, which creates larger voltage stress on converter components due to ringing and semiconductor output capacitance. The presence of leakage inductance has also led to the use of passive snubbers, active-clamping, and soft-switching. Additionally, transformers increase converter area, volume, weight, cost, and EMI. Although the development of bidirectional full-bridge converters are in high demand, half-bridge topologies have seen significant development, with their reduction in number of components. As a result, converters such as full-bridge primary and half-bridge secondary with synchronous rectification using coupled inductors have contributed a three times size reduction compared to its full-bridge counterpart. Lastly, wide band gap semiconductors are slowly being incorporated into EV DC-DC converters as they have better thermal conductivity, breakdown, and maximum operating junction temperature than silicon. Thus, SiC (Silicon Carbide) and GaN (Gallium Nitride) based semiconductors will reduce temperature issues, and have been used in hybrid vehicles like the Toyota Prius II.

The University of Wisconsin-Madison's Formula SAE team, Wisconsin Racing, published their Electrical Safety Form, documenting their solution to the DC-DC stage design. Wisconsin Racing utilized the Vicor Power DCM290P138T600A40 DC-DC module in conjunction with the LTC4417 PowerPath Controller to supply power to the Low Voltage system. With this DC-DC module, they were able to package a system with power flow control, pre-charge and discharge circuitry, a shutdown circuit, and a battery management system, in an

enclosure approximately 3.6 by 5 in. enclosure [7]. Unfortunately, information from other Formula SAE teams is not available. As such, most research is limited to general DC-DC converter designs. However, the DCM290P138T600A40 is also used by Cal Poly Racing for this year's car design. The Vicor DC-DC module boasts over 90% efficiency at 500 W from 200-400 VDC input with internal transformer in a 1 by 2 in. enclosure. The Vicor DC-DC achieved this with a 600 to 1000 kHz variable switching frequency, ZVS controller. However, the Vicor DC-DC converter is not a fully standalone controller and it requires 1 mF of external output capacitance and heat-sinks. Additionally, despite its high frequency operation, the Vicor DC-DC has approximately 400mV p-p of ripple [8].

Another potential DC-DC module available is the Cosel SNDBS700B12. Unfortunately, not much information can be gathered regarding the overall performance. The Cosel DC-DC is a 8.75 by 3.8 in. DC-DC converter with an efficiency of 90% from 200 to 400 V. An external heat-sink is also required for the Cosel DC-DC. Although limited information is available for the Cosel DC-DC, the given information implies that this solution is inferior to the Vicor DC-DC [9].

This project is to design, build, and test of a 500W, 200-400 VDC in to 14 VDC out, high efficiency, compact DC-DC converter. The Vicor DC-DC module is so much smaller than the RSP-500-12 that Cal Poly Racing cannot take full advantage of the packaging. Our DC-DC design will aim to produce a higher efficiency, lower output ripple DC-DC converter with a size between the Vicor DC-DC and the RSP-500-12 that doesn't require any external hardware.

3. Design Requirements

This chapter explains technical design requirements for the project. It encompasses those requirements related to electrical performance as well as physical dimension.

Technical Requirements

- **Input Voltage Range**

A 200-400 VDC input voltage range is a required design specification. The 88s8p LiIon battery configuration means that the tractive system voltage will vary from 220V to 369V during normal operation. Extending the range to 200-400V allows for a 20V safety margin.

- **Adjustable Output Voltage**

A 12-15V adjustable output voltage is needed to integrate with the low voltage battery. The low voltage battery used has its operating voltage vary from 11.2 to 14.4V. Cal Poly Racing does not expect the LV battery to drop to the minimum 11.2V state of charge, so a 12V minimum operating voltage is acceptable.

- **Control and Sense Ports**

DC-DC converter voltage and current data must be able to be constantly monitored to ensure safe operation. Cal Poly Racing also expects adjustable output voltage with external control.

- **Continuous Output Power Rating**

Cal Poly Racing expects to draw up to 400W of 12V power throughout the course of an endurance. Although the power draw is not continuous throughout the entire race, the

high power “bursts” can last for minutes, necessitating a continuous power rating of 500W.

- **Input/Output Overcurrent Protection**

Cal Poly Racing has a constant influx of new members. As such, assembly and operating errors occur that project leads are not able to fully prevent. Because of this, output short circuits are a possible event. In order to protect the DC-DC itself as well as the connected harness, overcurrent protection must be included in the design.

- **Input/Output Overvoltage Protection**

It is possible for an overvoltage condition to exist during regenerative braking, or if an incorrectly set voltage source is connected. The DC-DC must protect itself in order to prevent the need for purchasing a replacement DC-DC.

- **Input/Output Reverse Voltage Protection**

As mentioned previously, assembly and operational mistakes are possible in the DC-DC's lifespan. Another common error that occurs is reverse voltage events. These events are possible on the input and output of the device since energy sources in the form of batteries are connected on both sides. Because of this, the device must withstand input and output reverse voltage during a fault without permanent damage to ensure reliable operation.

- **Rated for Automotive Conditions**

Device undergoes rigorous vibrations as it is operated within a moving vehicle. Thus, most if not all components should be AEC-Q100 rated or better.

- **Isolated Design**

In accordance with 2019 Formula SAE Electric rules, EV.6.1.7, HV and LV on separate, clearly defined areas of the PCB, HV and LV areas are clearly marked, spacing between traces under conformal coating is more than 4mm. In accordance with EV.6.1.5, HV and LV within an enclosure is separated by moisture resistant, UL rated insulating materials rated for 150 degrees C or higher, and the spacing through air is greater than 30mm.

Additionally, at this power level, the design is isolated to prevent damage to low voltage components during a fault.

- **IP31 Rated**

In accordance with 2019 Formula SAE Electric rules, EV.6.5.3, the enclosure and connectors are protected from moisture with approximately an IP31 rating., meaning that the device is resistant to dripping water and greater than 2.5mm probes Although the rain test involves spraying water through a sprinkler directly into the car, The battery box enclosure that will contain the DC-DC converter will provide spray protection.

- **Small Volume**

Because Cal Poly Racing requires smaller components to improve vehicle packaging, the DC-DC converter must be roughly half the size of the currently used Meanwell RSP-500-15 DC-DC converter, which implies physical size less than 115mm x 127mm x 40.5mm.

- **Lightweight**

In order to be a usable replacement for the current module RSP-500-12, the DC-DC should be significantly lighter than the 2.866 lbs of the RSP-500-12. A target of less than 2 lbs total is a reasonable target.

- **Efficiency**

As a result of the previous requirements, converter's overall efficiency of at least 90% is required. The losses from the LV system is small compared to the total power consumption.

- **Line and Load regulations**

Because of the large voltage and load range required of the DC-DC converter, a strict line and load regulation is also required in order to keep the voltage from deviating too far away from the nominal value. A maximum regulation of 5% was selected for both line and load regulations.

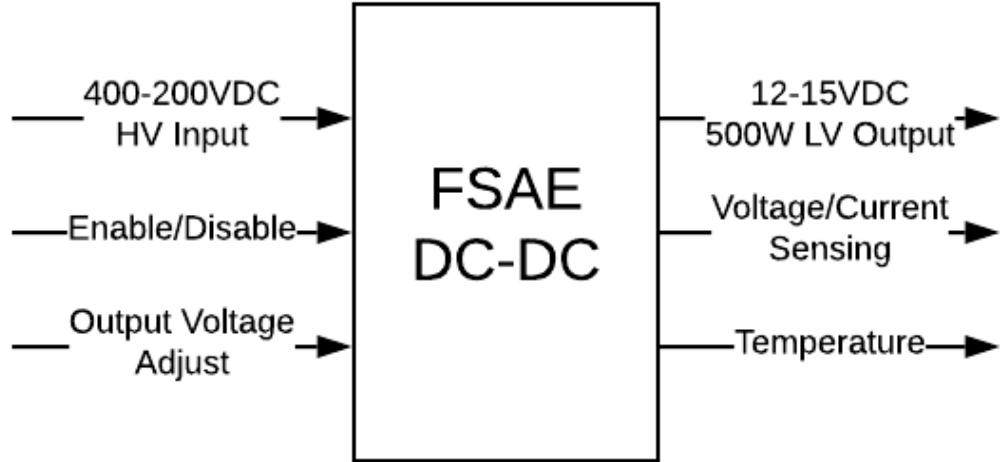


Figure 3-1: Level 0 Block Diagram

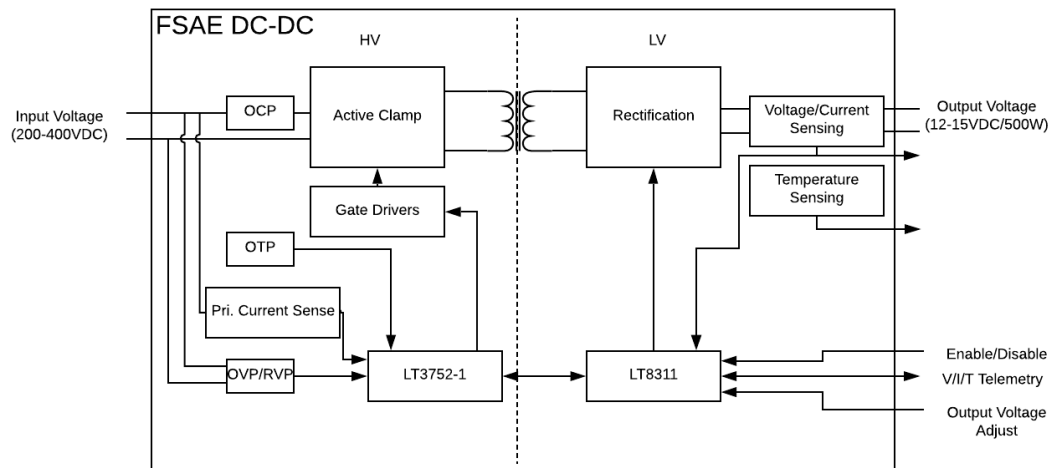


Figure 3-2: Level 1 Block Diagram

The Level 0 block diagram of the project shows that the proposed system consists of three inputs and three outputs. The three inputs contain the “high voltage” input, DC-DC converter enable control, and converter output voltage adjustment. The three outputs contain the “low voltage” output, voltage and current sensing signals, and various temperature signals.

The Level 1 block diagram of the project, as illustrated in Figure 3-2, consists of the main blocks on either side of the isolation. This includes the active clamp circuitry, various operating

protections, the two controller ICs, and various sensing blocks. Table 3-1 summarizes the design requirements for the project.

Table 3-1: Requirements List

Requirement	Value
Input voltage range	200-400V
Output Voltage Range	12-15V
Efficiency	> 90%
Additional Features	Control and Sense Ports
Power Rating	500W Continuous
Input/Output Protection	Overvoltage, Overcurrent and Reverse Voltage Protection
Ratings	Isolated, IP31, and Vibration Resistant
Size	< 115mm x 127mm x 40.5mm
Weight	2 lbs.

4. Design and Simulation Results

The Linear Technology LT3752-1 Active Forward Clamp Controller met all of the design requirements for the project. The IC is capable of ZVS and has a housekeeping controller. ZVS allows for improved switching efficiency, and the housekeeping controller reduces the size of the board by removing the need for an additional IC [6]. Additionally, the LT3752 has built-in integration with the LT8311 Synchronous Rectifier Controller with Opto-Coupler Driver for Forward Converters. This allows for synchronous rectification, which improves efficiency [7]. Excel calculators were made to determine component values.

Compared to other existing designs, our DC-DC converter uses a 300 kHz switching frequency, compared to the LT3752 Demo Board's 100 kHz switching frequency. This higher switching frequency allows us to achieve a comparable size to the demo board while carrying 2.5 times the power [8].

Output Inductor

$$L_{OUT} = \frac{V_{OUT}}{\Delta I_L \cdot f_{OSC}} \cdot \left(1 - \frac{V_{OUT}}{V_{IN}} \frac{N_p}{N_s} \right)$$

L_{OUT} = output inductance

V_{OUT} = output voltage

ΔI_L = output current ripple (peak-to-peak)

f_{OSC} = switching frequency

V_{IN} = input voltage

N_p = number of primary turns

N_s = number of secondary turns

Equation 4-1: Output Inductor Value Based on Output Current Ripple

The output current ripple was originally designed to be the recommended 40% of $I_{OUT(MAX)}$, but due to continuous and saturation current ratings of available inductors in our desired dimensions, the current ripple was increased. The increased current ripple should be compensated by the programmable slope compensation setting of the LT3752-1.

Primary MOSFET

Primary MOSFET selection depends on the maximum expected drain voltage and current.

Additionally, the power losses due to conduction, gate driver, and transition losses are used to determine the best fit for the converter.

$$V_{DS_{max}}(M1) = \frac{V_{IN}^2}{V_{IN} - V_{OUT} \cdot N}$$

$N = \text{ratio of number of secondary turns to primary turns}$

Equation 4-2: Maximum expected steady state drain voltage

MOSFET BV_{DSS} rating was at least 20% larger than the steady state V_{DS} value to account for tolerances in duty cycle, load transients, voltage ripple, and spikes caused by leakage inductance.

$$P_{M1} = P_{CONDUCTION} + P_{GATEDRIVER} + P_{TRANSITION}$$

Equation 4-3: Total Primary MOSFET Losses at Maximum Output Current

Several MOSFETs with different $R_{DS(ON)}$ and Q_G values were considered for the primary MOSFET, with the conduction loss being the primary loss. Additionally, the junction-to-case and junction-to-ambient coefficients were considered when determining heatsinks and overall package size. As a result, the LMG3410R050 Integrated GaN power stage was chosen for its low loss and low junction-to-ambient coefficient [9].

$$P_{CONDUCTION} = \frac{N_p}{N_s} \cdot \frac{V_{OUT}}{V_{IN}} \cdot \left(\frac{N_s}{N_p} \cdot I_{OUT(MAX)} \right)^2 \cdot R_{DS(ON)}$$

$I_{OUT(MAX)} = \text{maximum average output current}$

Equation 4-4: Primary MOSFET Conduction Losses

$$P_{GATEDRIVER} = Q_G \cdot V_{GS} \cdot f_{OSC}$$

$Q_G = \text{gate charge}$
 $V_{GS} = \text{gate-source voltage}$

Equation 4-5: Primary MOSFET Gate Driver Losses

Gate driver losses were not able to be calculated for the chosen MOSFET, the LMG3410R050, because of its fully integrated GaN design. Instead, the operating current value was used to calculate the gate driver loss.

$$P_{TRANSITION} = P_{TURN(ON)} = \frac{1}{2} I_{OUT(MAX)} \cdot \frac{N_s}{N_p} \cdot V_{DS} \cdot \frac{Q_{GD}}{I_{GATE}} \cdot f_{OSC}$$

Q_{GD} = gate to drain charge

I_{GATE} = gate current

Equation 4-6: Primary MOSFET Transition Losses

Transition losses only consist of turn-on losses, as it is assumed that ZVS is occurring on turn-off and will prevent any turn-off losses. The ZVS on turn-off is guaranteed, but depending on clamp switching timing, ZVS may be achieved upon turn-on, eliminating transition losses entirely.

Clamp MOSFET

Clamp MOSFET selection also depends on the maximum drain voltage and current. The main optimization goal is also to reduce $R_{DS(ON)}$ and Q_G , while meeting peak drain and body diode current ratings. Additionally, the maximum V_{DSS} is the same value as the primary MOSFET.

$$I_{MAG(MAX)} = \frac{1}{2} \frac{N_p}{N_s} \cdot \frac{V_{OUT}}{L_{MAG}} \cdot \frac{1}{f_{OSC}}$$

L_{MAG} = magnetizing inductance

Equation 4-7: Clamp MOSFET Maximum Drain Current

The maximum drain current of the clamp is significantly smaller than the primary MOSFET drain current and is double for safety margin due to tolerances in magnetizing inductance and transients.

$$I_{BD(MAX)} = \frac{N_s}{N_p} \left(I_{OUT(MAX)} + \frac{I_{L(RIPPLE)(P-P)}}{2} \right)$$

$I_{L(RIPPLE)(P-P)}$ = peak-to-peak output inductor current

Equation 4-8: Clamp MOSFET Maximum Body Diode Current

The pulsed body diode current occurs due to the active clamp topology, where the energy in the transformer's leakage inductance is conducted through the clamp body diode after the primary MOSFET turns off and before the clamp MOSFET is turned on.

Catch MOSFET

Catch MOSFET selection depends on the maximum drain-source voltage and current as well as maximum gate-source voltage. The main optimization goal is also to reduce $R_{DS(ON)}$ and Q_G .

$$V_{DS(MAX)} = V_{IN(MAX)} \cdot \frac{N_s}{N_p} \cdot Margin$$

Equation 4-9: Catch MOSFET Maximum Drain-Source Voltage

The catch MOSFET margin was chosen to be 1.5, and is typically between 1.5 to 2, to allow for a safety margin the $V_{DS(MAX)}$ equation. This accounts for voltage spikes from the leakage inductance of the transformer's secondary winding.

$$I_{D(RMS)} = \sqrt{(1 - D_{MIN}) \cdot \left(I_{LOAD(MAX)}^2 + \frac{I_{RIPPLE(P-P)}^2}{12} \right)}$$

Equation 4-10: Catch MOSFET Maximum RMS Drain Current

$$I_{OUT(PEAK)} = I_{LOAD(MAX)} + \frac{I_{RIPPLE(P-P)}}{2}$$

Equation 4-11: Secondary MOSFET Peak Current

The peak current, which is the peak inductor current, applies to both forward and catch MOSFETs.

Forward MOSFET

Forward MOSFET selection, like the catch MOSFET, depends on the maximum drain-source voltage and current as well as maximum gate-source voltage. The main optimization goal is also to reduce $R_{DS(ON)}$ and Q_G .

$$V_{DS(MAX)} = \frac{V_{OUT}}{1 - \frac{V_{OUT}}{V_{IN(MIN)} \cdot \frac{N_s}{N_p}}}$$

Equation 4-12: Forward MOSFET Maximum Drain-Source Voltage

MOSFET V_{DSS} rating was at least 20% larger than the calculated $V_{DS(MAX)}$ value due to energy being transferred between the magnetizing inductance and the clamp capacitor causing voltage “bowing”.

$$I_{RMS} = \sqrt{D_{MAX} \cdot \left(I_{LOAD(MAX)}^2 + \frac{I_{RIPPLE(P-P)}^2}{12} \right)}$$

Equation 4-13: Forward MOSFET Maximum RMS Drain Current

$$P_{loss(Q_G)} = V_{SUPP} \cdot (Q_{gCAT} + Q_{gFWD}) \cdot f_{OSC}$$

V_{SUPP} = supply voltage

Q_{gCAT} = catch MOSFET gate charge

Q_{gFWD} = forward MOSFET gate charge

Equation 4-14: Total Gate Charge Power Loss for Both Catch and Forward MOSFETs

Active Clamp Capacitor

Active clamp capacitance selection depends on the voltage ripple allowed by the components connected to SWP. Low clamp capacitance values will generate larger voltage ripple and thus, larger drain voltage on the primary MOSFET, but will require less magnetizing current.

$$C_{CL} = \frac{10}{L_{MAG}} \cdot \left(\frac{1 - D_{MIN}}{2 \cdot \pi \cdot f_{OSC}} \right)^2$$

Equation 4-15: Clamp Capacitance Value

$$C_S = 6 \cdot C_{CL}$$

Equation 4-16: Clamp Snubbing Capacitance

$$R_S = \frac{1}{1 - D_{MAX}} \cdot \sqrt{\frac{L_{MAG}}{C_{CL}}}$$

Equation 4-17: Clamp Snubbing Resistance

A snubber circuit will limit the peak voltages at the primary MOSFET drain during transients.

Mosfet Heatsinking

The packages for the Clamp Mosfet and the Catch Mosfet were TO-220 and required heatsinks to be chosen for them. The Primary Mosfet and the Forward Mosfets were surface mount and did not require any additional heatsinking beyond the recommended layout.

$$T_J = T_A + P_J * \theta_{JA}$$

T_J = Junction Temperature

T_A = Ambient Temperature

θ_{JA} = Thermal Resistance - Junction to Ambient in C/W

P_J = Mosfet Power Dissipation

Equation 4-18: Calculation of Ambient Temperature without Heatsinking

$$\theta_{HA-max} = \frac{T_{max} - T_A}{P_J} - \theta_{JA}$$

θ_{HA-max} = Maximum Thermal Resistance - Heatsink to Ambient in C/W

T_{max} = Max Desired Temperature

Equation 4-19: Calculation of Minimum Heatsink Performance

Active Clamp Switch Timing

In order to correctly perform ZVS for the primary MOSFET, AOUT, the clamp MOSFET gate signal, to OUT, the primary MOSFET gate signal, (t_{AO}) and OUT to AOUT (t_{OA}) delays need to be programmed. Both timings are generated from a single resistor R_{TAO} , which programs the clamp gate signal falling edge to the primary gate signal rising edge delay, or clamp MOSFET turn-off to primary MOSFET turn-on.

$$t_{AO(min)} = t_{AOUT(fall)} + t_{GDHL} + t_{ClpD(off)} + t_{Clp(fall)} - (t_{PriD(on)} + t_{Pri(rise)})$$

$t_{AOUT(fall)}$ = clamp gate signal fall time
 t_{GDHL} = gate driver high-to-low
 $t_{ClpD(off)}$ = clamp delay off time
 $t_{Clp(fall)}$ = clamp fall time
 $t_{PriD(on)}$ = primary delay on time
 $t_{Pri(rise)}$ = primary rise time

Equation 4-20: Minimum Timing from AOUT to OUT (Clamp Off, Primary On)

The minimum time between the clamp gate signal to the primary gate signal is determined using the chosen MOSFETs datasheet signal delay values. Having a larger t_{AO} value allows the main transformer's magnetizing current to discharge the primary MOSFET's drain voltage towards V_{IN} before turn-on.

$$R_{TAO} = \left(\frac{t_{AO} - 50ns}{3.8ns} \right) \cdot 1k\Omega$$

Equation 4-21: Switching Timing AOUT to OUT Resistor

The t_{OA} value is automatically generated and checked to make sure the clamp MOSFET does not turn on until the primary and forward MOSFETs are turned off to prevent shoot-through.

Synchronous Rectifier Timing

To synchronize the primary switching to the secondary side synchronous rectification, SOUT, the control signal to the secondary side, to OUT (t_{SO}), and OUT to SOUT (t_{OS}) need to be programmed for highest efficiency. Both timings are generated from a two resistors R_{TAS} , which programs the catch gate signal falling edge to the primary gate signal rising edge delay, or catch MOSFET turn-off to primary MOSFET turn-on, and R_{TOS} , which programs the primary gate signal falling edge to the catch gate rising edge delay, or primary MOSFET turn-off to catch MOSFET turn-on.

$$t_{SO(min)} = t_{SOUT(fall)} + t_{SyncD} + t_{CchD(off)} + t_{Cch(fall)} - (t_{OUT(rise)} + t_{PriD(on)} + t_{Pri(rise)})$$

$t_{SOUT(fall)}$ = sync signal fall time
 t_{SyncD} = synchronous transformer delay
 $t_{CchD(off)}$ = catch delay off time
 $t_{Cch(fall)}$ = catch fall time
 $t_{OUT(rise)}$ = primary rise time

Equation 4-22: Minimum Timing from SOUT to OUT (Catch Off, Primary On)

The minimum time between the catch gate signal to the primary gate signal is determined using the chosen MOSFETs datasheet signal delay values. The catch MOSFET should always be turned on when the primary MOSFET is turned off for continuous conduction. This results in very short non-overlap times between the SOUT and OUT gate signals.

$$t_{AS} = t_{AO} - t_{SO}$$

Equation 4-23: Minimum Timing from AOUT to OUT (Clamp Off, Primary On)

The t_{SO} delay is defined by the AOUT to OUT and AOUT to SOUT individual timings, so once the t_{SO} value is found, the t_{AS} value is determined to set the clamp off to primary on delay.

$$R_{TAS} = \left(\frac{t_{AO} - 50ns}{3.8ns} \right) \cdot 1k\Omega$$

Equation 4-24: Switching Timing AOOUT to SOOUT Resistor

$$t_{OS(min)} = t_{OUT(fall)} + t_{PriD(off)} + t_{Pri(fall)} - (t_{SOOUT(rise)} + t_{CchD(on)} + t_{Cch(rise)})$$

$t_{OUT(fall)}$ = primary signal fall time
 $t_{Pri(off)}$ = primary delay off time
 $t_{Pri(fall)}$ = primary fall time
 $t_{SOOUT(rise)}$ = sync signal rise time
 $t_{CchD(on)}$ = catch delay on time
 $t_{Cch(rise)}$ = catch rise time

Equation 4-25: Minimum Timing from OUT to SOOUT (Primary Off, Catch On)

Lastly, the t_{OS} is defined, which defines the primary MOSFET turn-off to catch MOSFET turn-on delay. This timing is programmed through the R_{TOS} resistor value.

$$R_{TOS} = \left(\frac{t_{AO} - 35ns}{2.2ns} \right) \cdot 1k\Omega$$

Equation 4-26: Switching Timing OUT to SOOUT Resistor

Simulation

Simulations were run many times over the course of the design in order to validate design equations, determine optimal values for parameters not covered by the datasheet, and retrieve data from hard to calculate expressions. The simulation depicted in Figure x is the final simulated schematic we used. This schematic was the test bed for all previous simulations. The schematic runs the converter through 5 load values, 0%, 25%, 50%, 75%, and 100% load, and with 3 different input voltage levels, 200V, 316V, and 400V. It should be noted that the schematic is not representative of the final design due to simulation issues. The first difference is

that most of the OC and other non-critical set resistors were not used because simulation errors sometimes resulted in a triggering of the various protection circuits that we do not expect to happen in actual operation. The second difference is that many of the mosfets are not representative of the actual mosfets we used. This is because most of the mosfets that we used did not have LTSpice models. The most notable of these is the primary mosfet, which has an auxiliary boost converter inside of it. As a result, the switch timing resistors were not implemented because the timings would not be representative of the actual system. Third, the input filter network is not included because our supply is an ideal voltage source in simulation and thus, would only slow down the simulation time. Finally, the gate driver is not present in the simulation. A gate driver LTSpice model was found, but the LTSpice solver was not able to reach convergence despite many simulation setting changes.

From simulation, we were able to confirm that the clamp and snubber network were still effective at 300 kHz, 500W. Additionally, we determined that a turns ratio of 4 and a primary inductance of 200 uH was the optimal combination for the main transformer. Deviating from this optimal point ended up in increased losses, due to the clamp network no longer performing optimally. Additionally, we found that reducing the current sense resistor on the main mosfet improved efficiency noticeably and improved operation at higher power levels. Finally, we were able to determine the average winding currents, which was critical information for determining the winding losses in the transformer. Components in the housekeeping supply, sync transformer and optoisolator were found to not significantly affect performance. The peak efficiency of our best simulation achieved 94.5% efficiency.

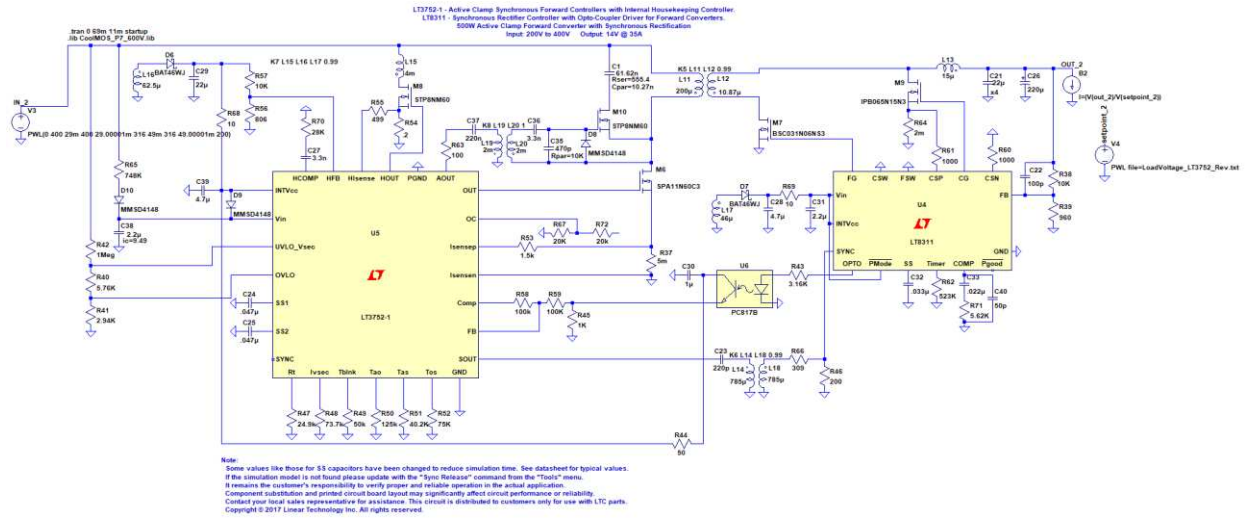


Figure 4-2: Final Simulation Schematic
 Source: Adapted from [10].

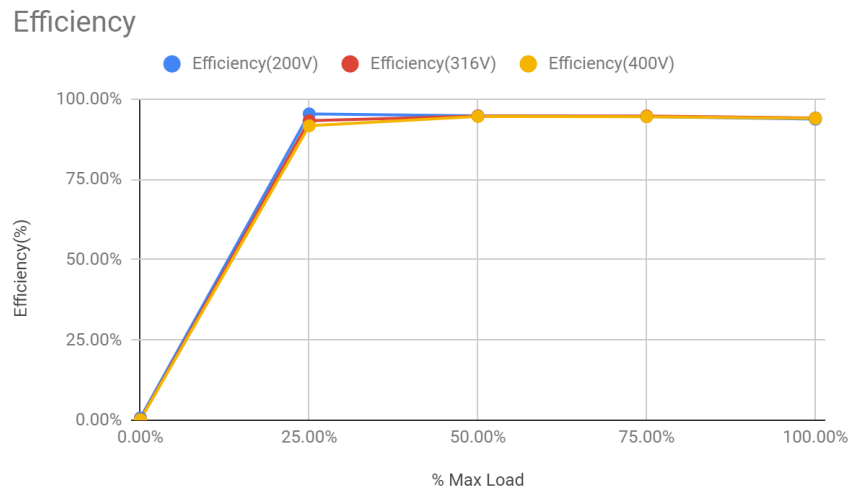


Figure 4-3: Efficiency Plot of Optimal Configuration

Transformer Design

The differing power and frequency between the demo board and our chosen design necessitated the design of a transformer. The transformer core was chosen using the area product method and the final design was finalized through a MATLAB script (see Appen. A).

The first step in designing a transformer is determining as many overall parameters as possible. Through simulation, we determined that we would be using a 300 kHz active clamp switching waveform, 400V max primary voltage, approximately 4:1 turns ratio, and 200 uH of primary side inductance. From here, the next step is to choose a core. A common method of selecting a core is through the area product method. The area product method looks at a core's cross-sectional core area and its cross-sectional winding area. This product can give a rough estimate of the total power handling of the core.

$$AP = A_w * A_e = \left(\frac{P_o}{K \Delta B f_T} \right) \text{cm}^4$$

P_O = Power Output in Watts

ΔB = Max Flux Density in Tesla

f_T = Transformer Operating Frequency

K = 0.014(Forward Converter Compensation Factor)

Equation 4-27: AP Product Guidance Equation

Source: Adapted from [11].

From here, we chose a core topology and size. We ended up choosing a N87 core material, ETD core (specifically, the B66358) from TDK. Choosing core topology will not be discussed further in the report. However, the ETD core was chosen due to the ease of winding and increased availability compared to PQ and other core types.

In the code, the first step is to calculate the skin depth. The skin effect is a phenomena in high frequency wires, where the flowing current induces an eddy current within the wire, causing the current density to increase around the edges of the wire, reducing the effective wire gauge. The skin depth is the depth where 30% of the current is present within that distance from the surface of the conductor.

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_r \mu_o}}$$

δ = skin depth

f = operating frequency

μ_r = relative permeability of conductor

μ_o = free space permeability

Equation 4-28: Skin depth equation

For a copper wire at 300 kHz, the skin depth is approximately 119 micron.

Since we started with a maximum allowable duty cycle, we are able to correlate that duty cycle to a maximum flux density, which was chosen to be 200mT. This flux density was chosen because it is the highest flux density that is characterized in the core loss plots for our chosen core.

$$N_{p1} = \text{ceil} \left(\frac{V_{in_{max}} * D_{max} * 1e4}{2 * B_{max} * f_t * A_e} \right)$$

N_{p1} = Number of Primary Turns

$V_{in_{max}}$ = Max Input Voltage

D_{max} = Duty Cycle in Decimal

B_{max} = Maximum Allowed Flux Density in Tesla

A_e = Transformer Core Cross-Sectional Area in mm²

Equation 4-29: Calculation of required primary turns

The actual operating flux density is determined by multiplying the max flux density for the chosen primary turns by the ratio of the operating duty cycle to the maximum duty cycle:

$$B_{op} = B_{max} * \frac{D_{op}}{D_{max}}$$

Equation 4-30: Calculation of Operating Flux Density

Now that we know the operating flux density, we can determine the core loss. TDK's website has core loss parameters for a given temperature and flux density, in units of mW/kg. These plots were inserted into the MATLAB script, interpolated, and multiplied by the core mass to determine loss. Lower flux densities will result in lower loss, which is why it is important to make the distinction between the maximum allowed duty cycle and the actual operating duty cycle.

Next, we must determine the core material and gapping required to achieve our desired primary inductance. In our research, most transformer designs are made without accounting for the transformer inductance, with the exception of designs for flyback supplies. However, our simulation indicated that failure to achieve a 200uH primary inductance would lead to decreased

efficiency and possible failure to operate. By changing the core material and gapping the core, we can reduce the effective permeability of the core enough to reduce the primary inductance to an acceptable value. A figure of calculating the inductance of a winding is the Al value. The Al value is in units of nH/N^2 , where N is the number of turns. Our core datasheet gave us the Al values for different cores and gaps. From here, we choose the gap that will give us an inductance that is closest to 200 μH .

Next is to calculate winding losses. From simulation data, we determined that 3.865A will flow through the primary windings on average during worst case, max power. Multiplying that current by the turns ratio will give us the secondary winding current. The length of the primary and secondary windings is determined by multiplying the number of turns by the mean path length, a parameter given by the bobbin datasheet.

From here there two potential methods for designing the transformer windings. The first method is to use single wire strands, the second is using Litz wire.

There are two different types of effects that increase resistance in transformer windings: skin effect and proximity effect. The skin effect is the increase in resistance caused by eddy currents causing the current to concentrate near the surface of the conductor. The proximity effect increase resistance in parallel wires due to interaction in their magnetic fields, which further crowds the current. The skin effect can be easily circumvented by using thinner gauge wire. On the other hand, the proximity effect can be reduced by interleaving the primary windings between secondary windings, since the secondary windings have currents that are out of phase with the primary windings, or by not paralleling wires. Because these are competing parameters and proximity effect losses are not easily calculable without EM simulation, the safest option is to use a small gauge wire to reduce skin effect losses and to interleave primary

and secondary windings until the all of the winding area is consumed. However, this will lead to a very difficult to manufacture transformer and the capacitance between the windings will be greatly increased.

When designing this way, it is important to make sure that the J, current density, between the primary and secondary windings are equal. This will ensure that the winding losses will be equal, which is the lowest loss point.

Litz wire is a bundle of electrically insulated wires that are twisted in a special way so every strand spends an equal amount of distance in every part of the cross-sectional area of the bundle. This allows Litz wire to have a larger effective diameter wire that bypasses the skin effect due to the thin wire and also fully use all of the wires in the strands because each wire is exposed to the same proximity effect forces at some point in the wire. This causes the initial resistance of the Litz wire to decrease as strand count increases. Because the proximity effect will increase resistance, the effective resistance of the litz wire will eventually start to increase as the number of parallel strands are added to the litz wire. Generally, the resistance of Litz wire is represented by F, which is the scaling factor of R_{ac} / R_{dc} . Taking the DC resistance of the Litz wire and multiplying it by F will give you the actual resistance of the Litz wire.

$$F = 1 + \frac{(\pi N_s N)^2 (d)^6}{192 \delta^4 b^2}$$

N_s = Number of Strands

N = Number of Turns

d = Individual Strand Diameter in mm

δ = skin depth in mm

b = Winding Width in mm

Equation 4-31: Calculation of F for Litz Wire

$$R_{ac} = F * R_{dc}$$

$$R_{dc} = \frac{l\rho}{\pi d^2 * Ns}$$

l = Total Wire Length

ρ = Copper Resistivity in $\frac{\Omega}{m^2}$

d = Wire Diameter in m

Ns = Number of Strands

Equation 4-32: Calculation of R_{ac}

Because R_{ac} will only be minimal over a very specific bundle outer diameter for a given strand diameter, and because only a limited combinations of Litz wire exist, it is not important to enforce an equal current density. Instead, it is better to test every combination of litz wires and interleaving, then picking the lowest loss combination. In our design, it was determined that no interleaving was required to reach the optimal litz wire solution.

While choosing different litz wire options, it is important to ensure that the total occupied area fits within the winding area. Winding area occupied is affected by the number of interleaving layers, the cross sectional area of the wire, and the number of turns. The relationship between the number of turns and the winding is why the area product equations exist. The larger the cross-sectional area of the core, the less turns are required for any given flux density, which increases the amount of winding area available for parallel wires. When using the litz wire, it is important to recognize that the actual area of the litz wire is does not scale linearly linearly with the number of strands. Additionally, we must take into account the wasted space between the round cavities when laying wires together. This was accounted for by increasing the wire area by 25%.

$$A_{litz} = \pi * (1.1 * (1.28 * \sqrt{N_s} * d))^2$$

Equation 4-33: Empirical Formula for Calculating Litz Cross-Sectional Area

The MATLAB script takes all of these equations and uses them to find the lowest loss solution. First, the script creates an array of different possible B. For each different B, we calculate the core loss, then try every combination of litz wire and interleaving to find the best winding combination that will fit within the winding area. With this information, we have the best possible total loss in the transformer for a single B. As we iterate through the different B's in the array, we keep track of the losses. When the loop completes, the script selects the B and winding combination with the lowest loss.

Final Design

Figures 4-x through 4-x represent the final design of the DC-DC converter. All component values calculated through the previous simulation and design equations are used in the schematic. Additionally, test points, reverse voltage and overcurrent protection, jumpers, and an enable pin are included. When power is not supplied to the enable pin, it blocks the supply voltage to the LT3752 undervoltage lockout pin in order to force the converter into shutdown pin. The jumpers allow the user to force the DC-DC to be off through the same method as the enable pin, or disable the overcurrent protection. During simulation overcurrent protection was occasionally an issue, so a disable was added in the event that false overcurrents were occurring. The reverse voltage protection and overcurrent protection are achieved through a fuse-diode circuit. Under normal conditions, current flows through the fuse and the diode is reverse biased. In the event of an overcurrent, the fuse blows, protecting the DC-DC. In the event of a reverse voltage, the diode becomes forward bias and short circuits the supply through the fuse. This causes the fuse to blow, disconnecting the circuit from ground.

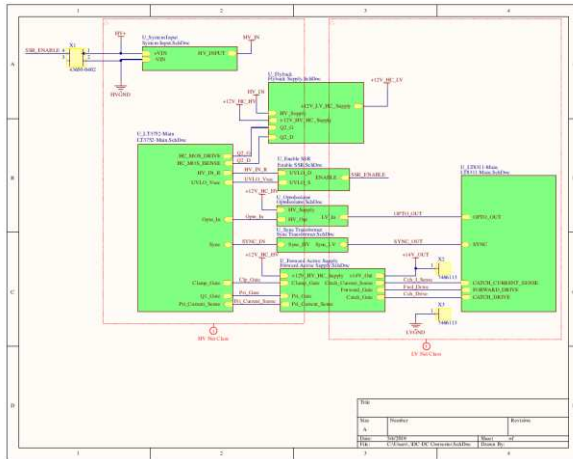


Figure 4-4: DC-DC Top Schematic

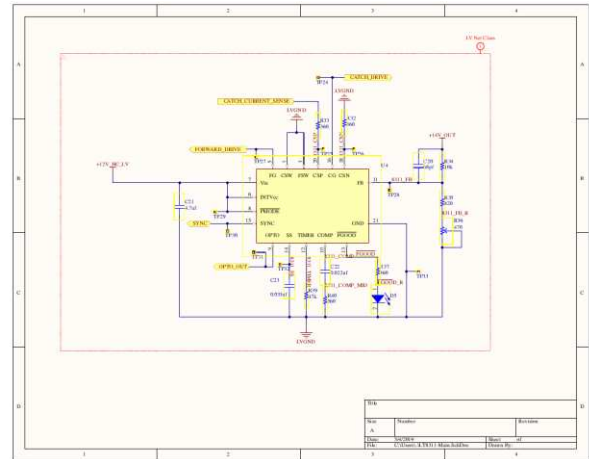


Figure 4-5: LT8311 Schematic

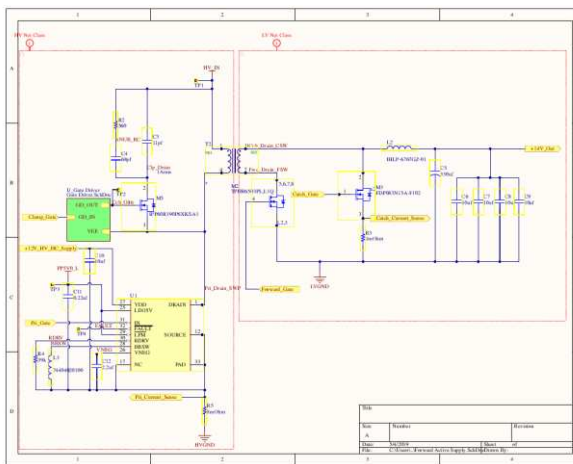


Figure 4-6: Forward Active Schematic

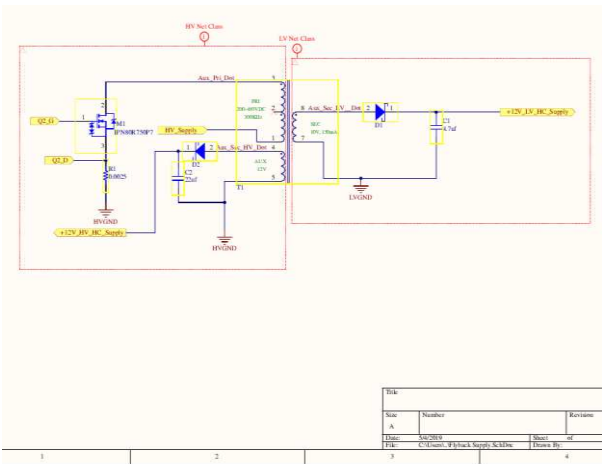


Figure 4-7: Flyback Schematic

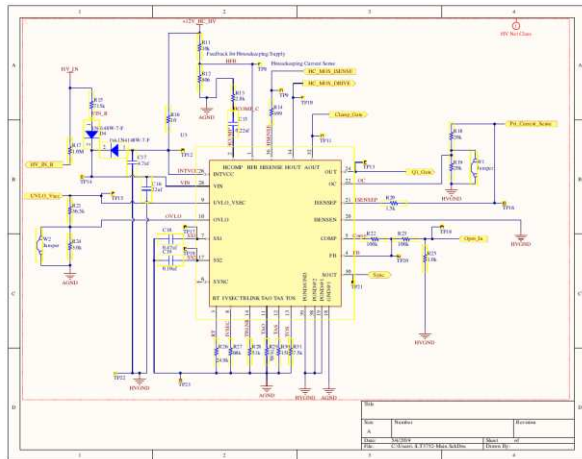


Figure 4-8: LT3752-1 Schematic

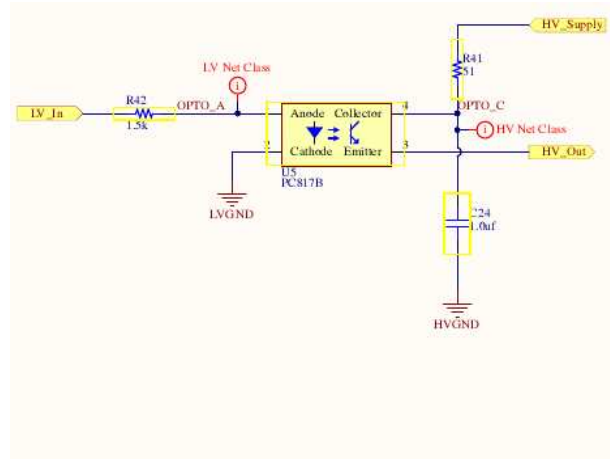


Figure 4-9: OptoIsolator Schematic

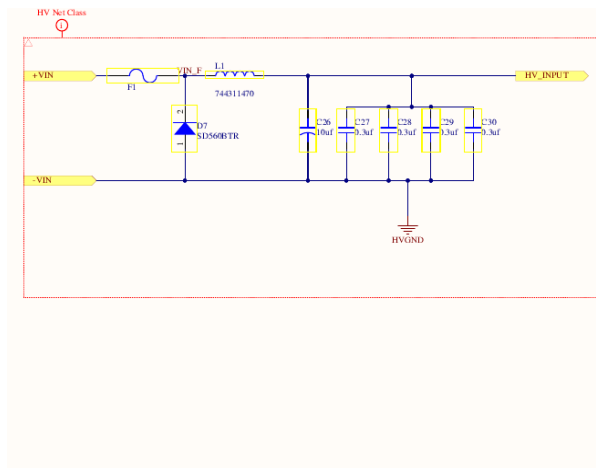


Figure 4-10: Input Fusing and RVP Schematic

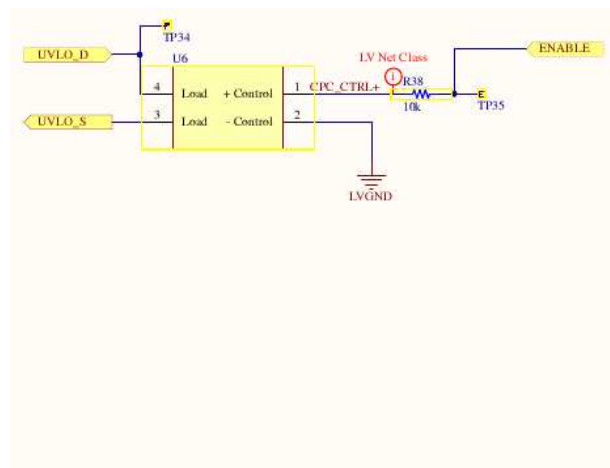


Figure 4-11: DC-DC Disable Schematic

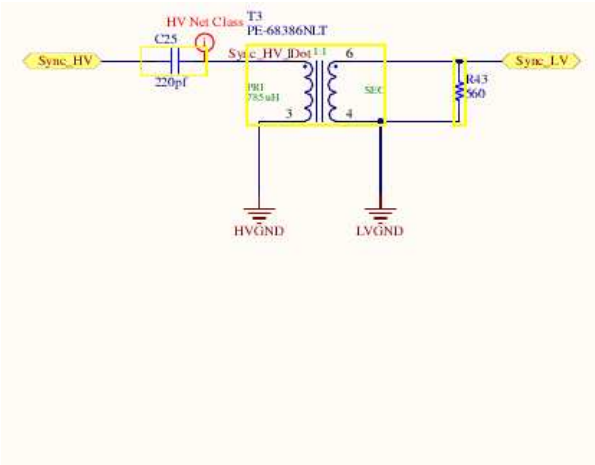


Figure 4-12: Sync Transformer Schematic

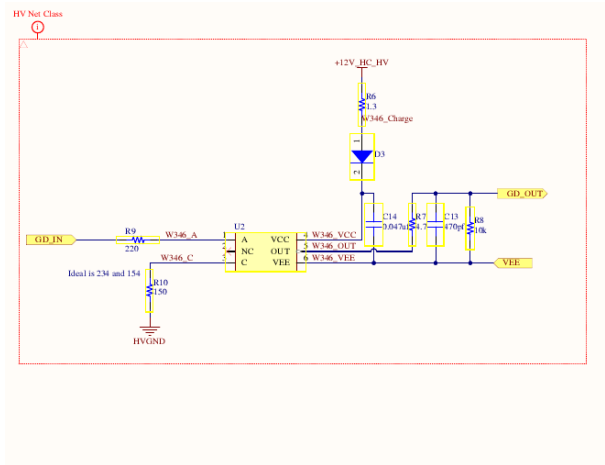


Figure 4-13: Gate Driver Schematic

5. Hardware Test and Results

The construction of the DC-DC is separated into two parts: the construction of the transformer, and the main board.

After purchasing the required transformer materials that were chosen in Chapter 4, we began winding the transformer. The primary windings were the inner layer and the secondary on the outer layer. As mentioned in Chapter 4, there was no need for interleaving. The inner layer was used for the primary winding in order to reduce leakage inductance on the primary side. Additionally, the gapping needed to achieve the correct leakage inductance will cause the inner layer to experience higher proximity effect resistance. Therefore, it is better to have the higher current, secondary winding further away from the center leg of the core since the secondary winding experiences higher currents.

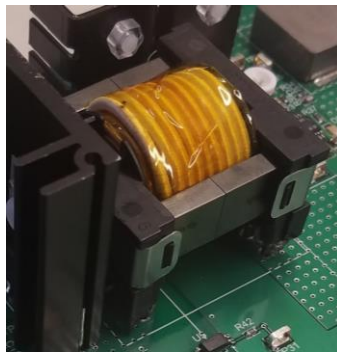


Figure 5-1: Final Constructed Transformer

While winding the transformer, we tried two separate methods for winding the transformer, which we called low leakage, and crossover. The low leakage winding method reduces the total leakage inductance while the crossover method lowers the maximum voltage between adjacent windings. As seen in Table 5-1, there are no significant changes in R_{ac}

between the two methods; therefore, we chose to follow the crossover winding method to reduce the risk of arcing in the transformer.

The primary side inductance matched calculations very well. The chosen 0.2mm gap gave us a primary side inductance of approximately 454uH, which is close to the predicted 400uH. Because the datasheet did not provide data on gaps formed by combinations of different gaps, we tested all the different combinations of gaps and found that a 0.4mm total gap gave the closest primary side inductance to 200uH. On the other hand, Rac measurements deviated significantly from the calculated data. Most Rac measurements were found to be over 30 times greater than expected.

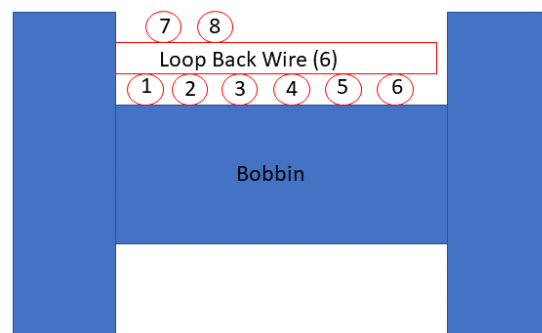


Figure 5-2: Low Leakage Winding Depiction

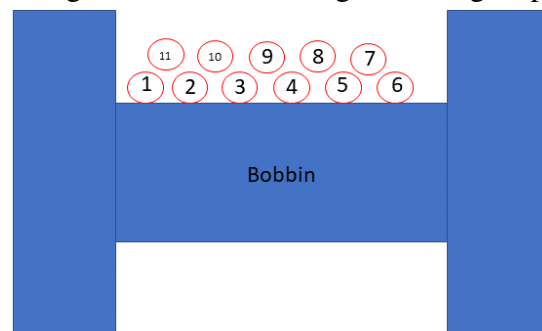


Figure 5-3: Crossover Winding Depiction

Table 5-1: Transformer Design Test Data

Primary Winding Gap	L@300kHz (uH)	Rac@300kHz	Rdc (mOhm)	Winding
.4mm gap	268.8	2.2	Not Measured	Lleak attempt
.3mm gap	334	2.436	Not Measured	Lleak attempt
.2mm gap	454.6	3.097	Not Measured	Lleak attempt
.1mm gap	721	5.3	Not Measured	Lleak attempt
0 gap	2.5	123	Not Measured	Lleak attempt
.4mm gap	269	2.53	50.93	Crossover
.3mm gap	336.3	2.77	50.93	Crossover
.2mm gap	454.2	3.6	50.93	Crossover
.1mm gap	709	6.8	50.93	Crossover
0 gap	2.49	124.4	50.93	Crossover
Secondary Winding				
.4mm gap	21.09	0.3439	4.065	
Calc'd N1/N2	3.571394703			

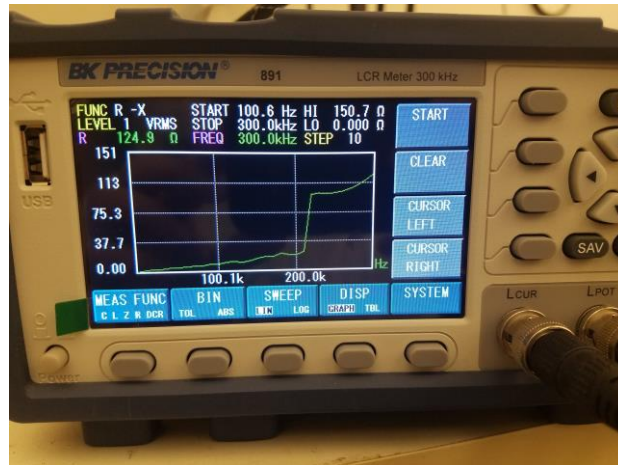


Figure 5-4: Transformer Measured Primary Side AC Resistance

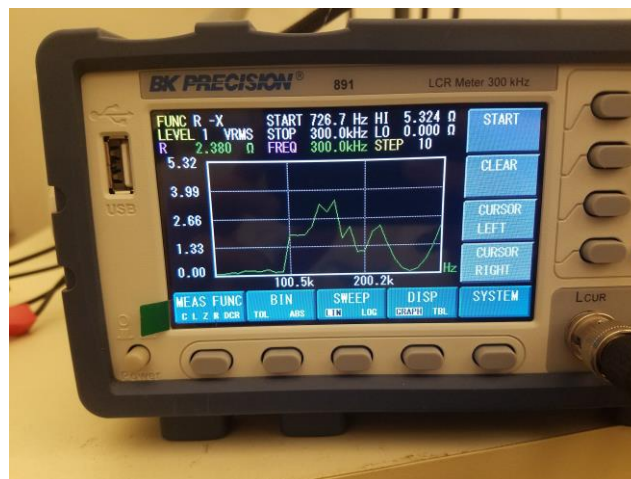


Figure 5-5: Transformer Measured Secondary Side AC Resistance

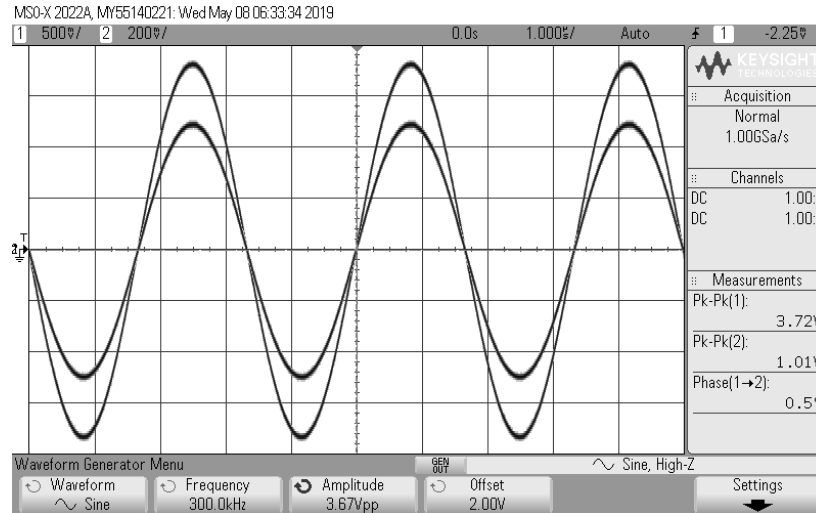


Figure 5-6: Transformer Turns Ratio Validation

After AC resistance measurements were made, we validated the turns ratio by inputting a sinusoid and measuring the output sinusoid amplitude as shown in Figure 5-6.

The DC-DC converter main board was laid out in Altium Designer, layout and board shown in Figures 5-7, 5-8, and 5-9. Decoupling capacitors were used in close proximity to each IC's V_{IN} and GND. The T_{AO} , T_{AS} , T_{OS} , T_{BLNK} , $IVSEC$, and RT resistors were placed close to each pin of the LT37352-1. A single, closed off AGND plane was used to ground these resistors to pin 18 of the LT37352-1. Kelvin connections were used to connect the primary current sense resistor for the LT37352-1 to the I_{SENSEP} and I_{SENSEN} pins. The current sense resistor for the housekeeping supply also had its ground connection close to PGND, pin 38. The gate driver traces, HOUT, AOUT, SOUT, and OUT were made as short as possible, bringing the MOSFETs closer to the LT37352-1. High current traces were expanded and connected on multiple layers with vias to reduce current path resistance. Additionally, the low voltage and high voltage portions of the PCB were separated by at least 4 mm to comply with Formula SAE rules [12]. The board was designed for a 1 oz, 4 layer PCB. Once the layout was finished, the PCBs were

etched by Bay Area Circuits. To solder the board, homemade stencils were created to decrease manufacturing time. The base stencil material consisted of two layers of Kapton tape stuck on either side of a piece of a paper. The Kapton-paper composite was then cut on a laser cutter using the paste layer from the Altium board layout, forming a makeshift stencil. Solder paste was applied and all non-through hole components on the top layer were placed and reflowed using a microscope and toaster oven. The remaining through-hole and bottom layer components were soldered by hand. For the through-hole MOSFETs, their heat sinks were bolted on using plastic bolts and nuts with insulating thermal pads.

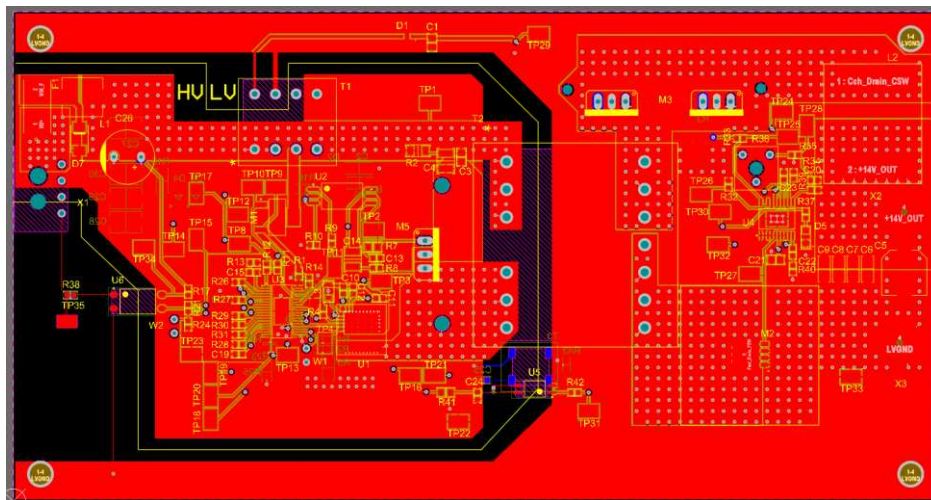


Figure 5-7: DC-DC Main Board Layout

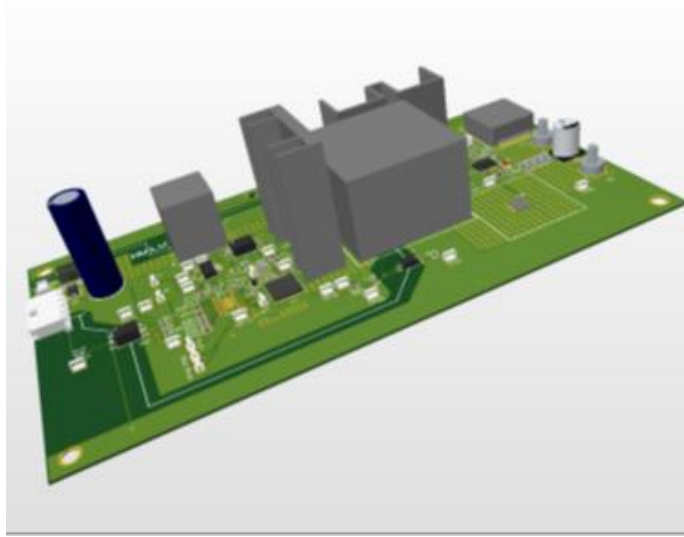


Figure 5-8: DC-DC 3D Render

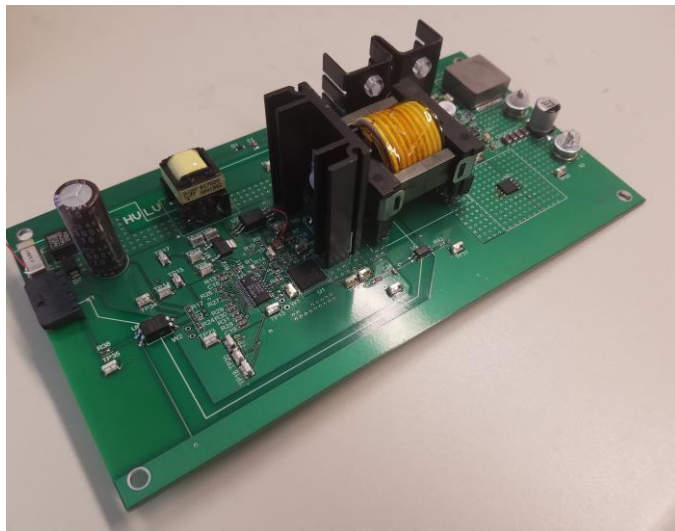


Figure 5-9: DC-DC Final Constructed Board

To sufficiently monitor all relevant signals and rails of the converter, multiple oscilloscopes and differential probes were used to measure and characterize the converter. The basis of the test setup consisted of monitoring the input and output voltage and current. The final test setup consisted of two 600V, 1.5A power supplies in parallel, sourcing the converter as shown Figure 5-10. The HV current path was conducted through an ammeter, and eventually to

the DC-DC converter. Additionally, the input voltage of the DC-DC converter was monitored using a multimeter after the input fuse of the converter to indicate whether or not the converter had passed its overcurrent limit. The output voltage was monitored using another multimeter, and the output current was measured using a BK Precision electronic load when the converter was loaded.

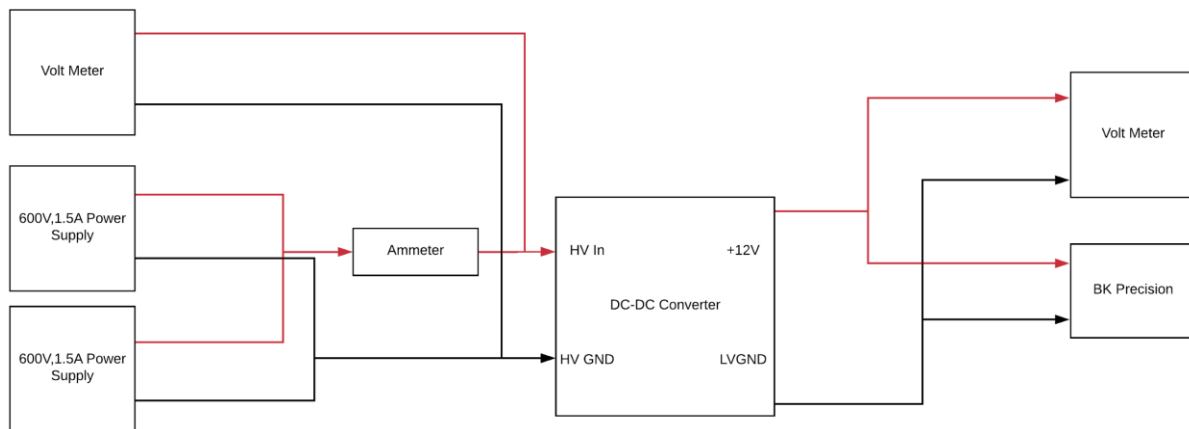


Figure 5-10: DC-DC Multimeter, Supply, and Load Block Diagram

As for the signals and power rails of the converter, three different oscilloscopes were used and synchronized to measure all the relevant signals. However, the ground prong of each oscilloscope connected the chassis together and thus, the grounds of each oscilloscope probe were connected together. Because of the nature of isolated power supplies, signals on the high voltage and low voltage sides of the converter could not be normally measured by neither the same oscilloscope, nor multiple oscilloscopes. To measure the signals on both sides of the converter, the signals on one of the sides needed to be measured differentially, with probes from all three oscilloscopes grounded to the same point. In this case, all probes from all three oscilloscopes were grounded to the high voltage ground. This meant that high voltage signals could be measured normally, while all low voltage signals needed to be measured differentially.

The high voltage signals were measured using Rigol 1, as shown in Figure 5-11. In order to use probes efficiently, only one probes from the Rigol 2 and Keysight oscilloscopes were used to measure the low voltage ground signal. A Matlab script was used to perform math on the measured data to subtract the Rigol 2 and Keysight signals from their measured ground signal to extract the signal waveforms after the fact. Additionally, this Matlab script synchronized the data measured from each oscilloscope to the same time scale. To accomplish this, each oscilloscope used the same time division scale and were externally triggered to each other. In total, 7 signals, (9 including low voltage grounds), were measured, including OUT (primary MOSFET gate signal), AOUT (clamp MOSFET gate signal), SWP (primary MOSFET drain voltage and bottom leg of primary side of the transformer), FSW (forward MOSFET drain voltage and bottom leg of secondary side of the transformer), CSW (catch MOSFET drain voltage and top leg of secondary side of the transformer), FG (forward MOSFET gate signal), CG (catch MOSFET gate signal), and HV IN (high voltage input and top leg of the primary side of the transformer).

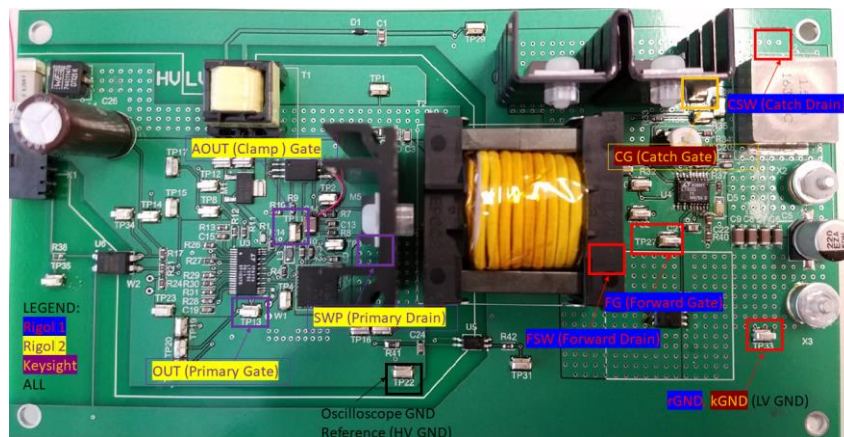
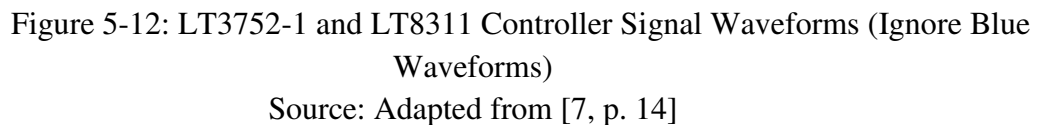


Figure 5-11: DC-DC Oscilloscope Measurement Point Setup

Input Voltage	Load	Efficiency	Output Voltage
200V	0%	N/A	9V
200V	25%	93%	7V



According to Figure 5-12 from the datasheet, SWP and FSW mirror each other with FSW being stepped down by the turns ratio. This is accomplished by transformer action on opposing legs of the transformer. Additionally, CSW, the top leg of the secondary side, resembles a square wave with ringing at the rising edge.

Table 5-3: DC1929A Demo Board Waveforms¹

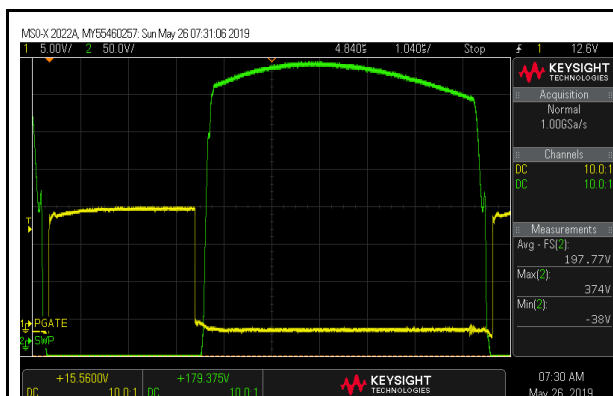


Figure 5-13: DC1929A OUT and SWP

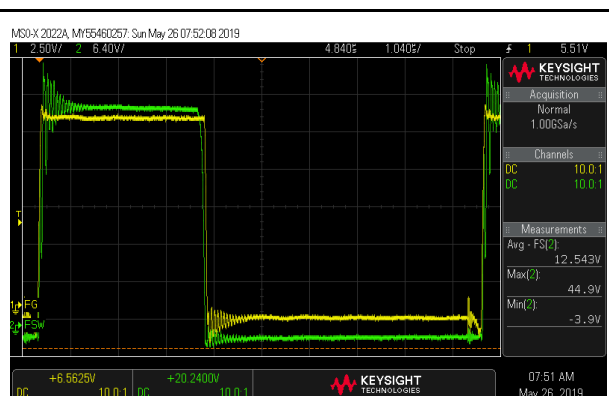


Figure 5-14: DC1929A FG and FSW

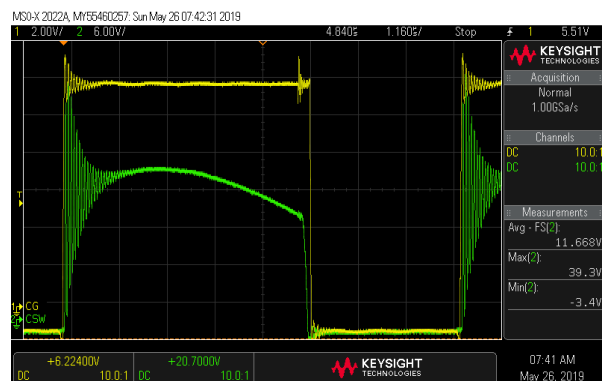


Figure 5-15: DC1929A CG and CSW

¹ All signals tested using 316V input at 2A load.

The DC1929A Demo Board was tested to serve as example control signals of how the DC-DC converter should look. As shown in Figure 5-14, the FSW waveform closely resembles a square wave with ringing on the rising and falling edge. The voltage steps shown in the datasheet waveforms are small, and not clearly seen in the SWP and FSW waveforms in Figures 5-13 and 5-14. However, both waveforms roughly resemble a square wave.

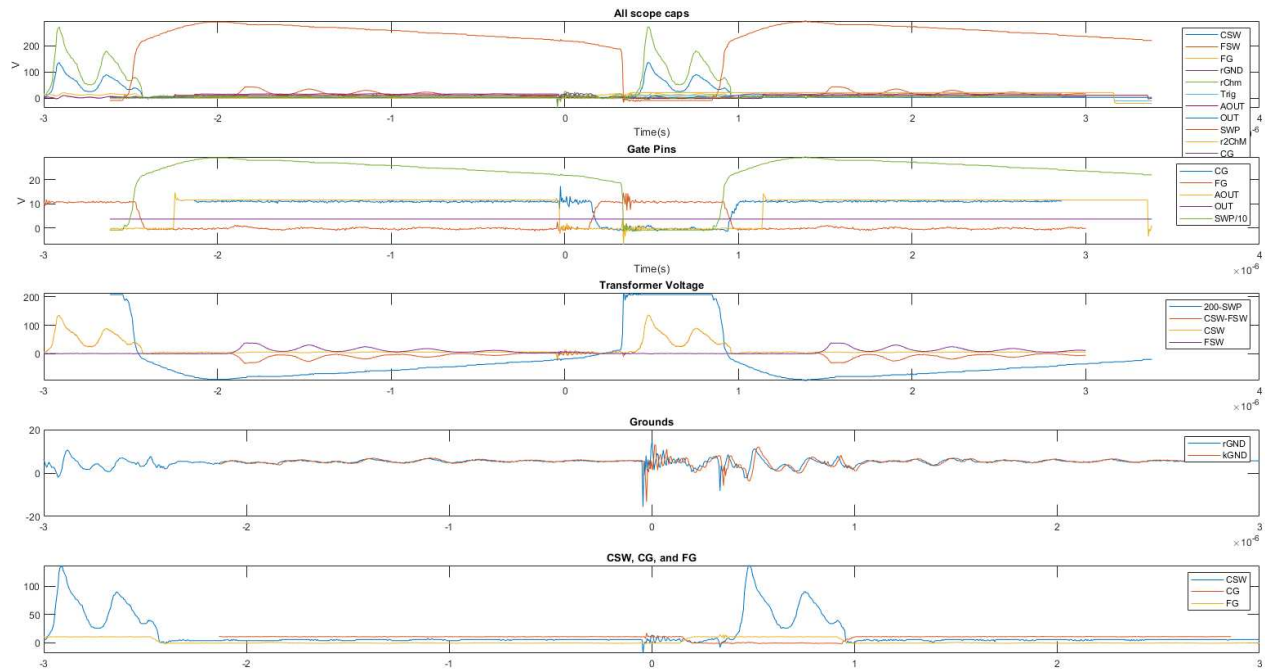


Figure 5-16: 8A load test with ringing

Figure 5-16 indicates the major problem that occurred while testing the DC-DC converter. As seen in the 3rd from top plot, the primary and secondary voltage of the transformer do not match. This does not match expected transformer operation. Additionally, the CSW and FSW waveforms have multiple voltage spikes, likely from ringing. This is the result of the mismatched transformer operation, as the SWP signal is very close to a square wave while the FSW signal is ringing.

6. Conclusion

This project entails the design and construction of the HV (High Voltage) to LV (Low Voltage) DC-DC converter for the Formula SAE Student Design Competition (FSAE) electric vehicle car. The final DC-DC converter was able to demonstrate successfully a high frequency power transformer design, which further provides a useful framework to assist in the speed of development of future DC-DC projects. There were some issues encountered during the testing of the converter, which include meeting the load specifications due to excess ringing on the primary and secondary side leading to eventual failure of the power MOSFETs during light load operation. Additional learning experience gained from the project is the knowledge needed for proper heatsink selection and the balancing of switching and conduction losses to improve efficiency. Finally, we also developed a measurement system to allow for the synchronization of multiple oscilloscopes for differential probe measurements.

Future improvements falls into two categories: improvements that could help the DC-DC meet specifications, and improvements that should be implemented after meeting specifications.

Improvements focused on helping the DC-DC meet specifications are centered around fixing the ringing, as we believe it is the primary source of component failure and heating in the DC-DC. The intensity of the ringing is potentially due to the measurement setup. Using high voltage differential probes would significantly decrease the complexity of the measurement setup by reducing the amount of probes required. Additionally, the parasitic capacitance of the oscilloscope probes should also be reduced by using differential probes. Another source of resonance is through the input capacitance of the secondary side MOSFETs.

The frequency of the ringing also decreased after increasing the output inductor value, so having a larger output inductor may help as well. In the Demo Board results, the voltage at the top leg of the transformer resembled the top half of a sine wave with significant ringing. This may indicate that a higher frequency resonance is superimposed on a lower frequency resonance, with the higher frequency resonance being switching or leakage inductance related, and the lower frequency resonance being related to the resonance between the output inductor and output capacitance.

As another improvement, the transformer needs more testing and validation. Because we lacked a proper test fixture, we were unable to test our transformer for leakage inductance, power handling and load regulation capabilities. Transformer characterization is important because high secondary side leakage inductance could be the reason why the ringing is only present on the secondary side. Designing an H-bridge or fixed timing active clamp forward converter would help characterizing the source of the ringing.

Improvements focused on improving a functional DC-DC center on performance improvements and quality of life changes. One improvement includes reducing the overall size of the converter in the width and length dimensions. These improvements will mainly decrease the power path length, reducing resistance, and benefit packaging for future Formula SAE Electric vehicles.

In terms of improving our ability to debug the converter, the clamp MOSFET should be moved or become a surface mount package, which would assist the removal of the transformer core during board reworks. Additionally, the clamp MOSFET heatsink was oversized and can be reduced. Several nets of converter were missing that would benefit debugging should be added. Some of these nets include the HV net before the input filter, the output voltage, and primary

MOSFET pins. The output connector system bolting wire lugs to M3 screws was tedious to connect and remove and should be altered to improve removal.

In order to further reduce size, water cooling could be included to decrease the required cooling effort. This would allow designers to use smaller switching IC's, which should improve efficiency, as the switching energy will decrease. This would also decrease the total required mass for heatsinks.

Another potential improvement is the addition of a complete microcontroller system. This would significantly improve the tunability of the design as well as the amount of telemetry that could be sent back. In this project, voltage and current sensing were not able to be easily transmitted to an external processor. Additionally, the output voltage could only be adjusted manually. Both of these operations could be accomplished using a microcontroller system.

7. References

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8. Appendix

Code for Transformer Design

Transformer.m

```
%% Initializations
clearvars
load('wireSizes.mat');      % Loading wire gauge tables
load('litzOptions.mat');    % Loadings Litz wires of form (Nstrand,
Gauge)
% Litz wires available from
https://www.mikeselectronicparts.com/litz-wire/
Bdes = (0.005:0.001:.2)';  % Desired Flux Swing (Arbitrary Starting
Value)
K = 0.014;                  % Compensation Factor

%% Constants
rho_cu = 1.68e-8;           % Copper Resistivity (ohm m)
%% Desired Values
Lpri_des = 200e-6;          % Desired Primary Inductance
TR = 4;                     % Desired Turns Ratio
Jdes = 2.2;                  % Desired Current Density (A/mm^2)
eta = 0.3;                  % Expected Efficiency
DutyLim = 0.75;             % Max Duty Cycle
DutyOp = 0.3;               % Operating Duty Cycle
Ninter = 4;                 % Max interleaving layers
T = TransformerModel2(500, 300e3, DutyLim, DutyOp, 400,200, 0.75);
AP = (T.Po./(K.*Bdes.*T.ft)).^(3/4); % Estimated Area Product
Required (cm^4)
%% Core Properties
% B66358
T.Core = Core(0.93, 70.4, 76, 5350, 97, 19.4, 52.8, 28,...
    [1470 1610 1670 457 281 148 91], 0.3, ...
    [2000 2200 2250 621 383 201 124],...
    [0.025, 0.050, 0.100, 0.200;
    0.00333, 0.014433 0.084536 0.500732],...
    [0 0 0 .1 .2 .5 1]);
```

```

T.wPri = LitzWire();
T.wSec = LitzWire();
%% General Design Calculations
%T = V*s/m^2
Np1 = ceil(T.VinMax .* T.Dlim.* 1e4
./(2.*Bdes.*T.ft.*T.Core.Ae*0.01));
Bmact = T.VinMax .* T.Dlim.* 1e4 ./ (2 .* Np1 .* T.ft .*
T.Core.Ae*0.01); % Actual Max Allowed B
Bact = Bmact.*T.Ddes./T.Dlim; %Getting Actual used B
Np2 = ceil(Np1/TR);
TR_act = Np1./Np2;
Al_req = Lpri_des ./ Np1.^2 *10^9;
% Determining the correct Al for the job
gChosen = zeros(1,length(Al_req));
AlChosen = zeros(1,length(Al_req));
LChosen = zeros(1,length(Al_req));
for i=1:length(Bdes)
    Altemp = T.Core.Al(T.Core.Al > 1.2.*Al_req(i));
    muiChosen = T.Core.mui(length(Altemp));
    gChosen(i) = T.Core.g(length(Altemp));
    AlChosen(i) = Altemp(end);
    LChosen(i) = AlChosen(i) * Np1(i)^2*1e-9;
end
clear Altemp
T.Ipri = 3.865; % Measured Values
Isec = T.Ipri*TR_act;
LenPri = Np1 * T.Core.LN * 0.001; % Primary Wire
Length in m
LenSec = Np2 * T.Core.LN * 0.001;
Pcore = T.Core.getCL(Bact);
A = T;
A = repmat(A,1,length(Bdes));
for Bindex = 1:length(Bdes)
    T.Np1 = Np1(Bindex);
    T.Np2 = Np2(Bindex);
    T.Isec = Isec(Bindex);
    T.Bop = Bact(i);
    T.Lpri = LChosen(i);
    T.Core.gOpt = gChosen(i);

```

```

    T.Core.AlOpt = AlChosen(i);
    T.TR = TR_act(i);
    T.Pcore = Pcore(i);
    T.Bmax = Bmact(i);
    A(Bindex) = T.findLitz(LitzOptions, LenPri(Bindex),
    LenSec(Bindex), ...
        Ninter);
end
% Finding Optimal Operating Point
[val, i] = max([A.eta]);
T = A(i);
%% Plots
figure(1)
plot(Bdes,[A.eta])
xlabel('Bmax')
ylabel('Efficiency(%)')
title('Efficiency vs. Max B')

```

TransformerModel.m

```

classdef TransformerModel2
    %TRANSFORMER Holds all of the relevant information to the final
    %transformer design
    % Detailed explanation goes here

    properties
        %% Objects
        Core          % Model of Transformer Core
        wPri           % Primary Wire Model
        wSec           % Secondary Wire Model
        %% Vars
        AwPri          % Area used by primary windings
        AwSec          % Area used by secondary windings
        Bmax           % Max B used
        Bop            % Nominal B used
        Dlim           % Max Possible Duty Cycle
        Ddes           % Desired Duty Cycle
        eta            % Efficiency
        ff             % Winding Fill Factor
    end
end

```

```

ft          % Transformer Operating Frequency
Ipri        % Primary side peak current
Isec        % Secondary side peak current
JPri        % Current density of the Primary
JSec        % Current Density of the Secondary
Lpri        % Primary side inductance
Ninterleave % Number of interleaving section PER PHASE
Np1         % Primary Number of Turns
Np2         % Secondary Number of Turns
Po          % Output Power in Watts
Pcu         % Power Dissipated in windings
Pcore       % Power Dissipated in the core
PcuPri      % Power Dissipated in the primary windings
PcuSec      % Power Dissipated in the secondary windings
Ptot        % Total Dissipated Power
sd          % Skin Depth
TR          % Turns ratio
VinMax      % Max Primary Voltage
VinMin      % Minimum Primary Voltage

end

methods
    function obj = TransformerModel2(Po, ft, Dlim, Ddes, VinMax,
VinMin,...
                                ff)
        %UNTITLED3 Construct an instance of this class
        % Detailed explanation goes here
        obj.Po = Po;
        obj.ft = ft;
        obj.sd = getSkinDepth(obj.ft);
        obj.Dlim = Dlim;
        obj.Ddes = Ddes;
        obj.VinMax = VinMax;
        obj.VinMin = VinMin;
        obj.ff = ff;
    end
    function obj = findLitz(obj,LitzOptions,lenPri,lenSec, N)
        litzWires = LitzWire(LitzOptions, [obj.Np1, obj.Np2],
obj.Core.Ww, obj.sd, [lenPri,lenSec]);

```

```

        Ninterleaves = (1:1:N);    %Number of interleaved blocks
per side (4 primaries)
        AwPriM =
repmat(reshape(.01:.01:(obj.Core.Aw*0.8*obj.ff/obj.Np1),1,1,[])...
        ,1,length(Ninterleaves),1)./Ninterleaves;    %List of
optional Aw's
        AwSecM = (obj.Core.Aw*0.8*obj.ff -
AwPriM.*Ninterleaves.*obj.Np1)/obj.Np2./Ninterleaves;
        RPriAC = repmat(litzWires.getRac(1')./Ninterleaves, 1, 1,
length(AwPriM));
        RsecAC = repmat(litzWires.getRac(2')./Ninterleaves, 1, 1,
length(AwSecM));
        RPriAC(AwPriM - ...
        repmat(Ninterleaves.*[litzWires.Ao]', 1, 1,
length(AwPriM)) < 0) = NaN;
        RsecAC(AwSecM - ...
        repmat(Ninterleaves.*[litzWires.Ao]', 1, 1,
length(AwSecM)) < 0) = NaN;
        PpriAC = obj.Ipri^2.*RPriAC;
        PsecAC = obj.Isec^2.*RsecAC;
        [obj.Pcu, indexAw] = min(min(min(PsecAC)+min(PpriAC)));
        [~, indexInter] = min(min(PsecAC)+min(PpriAC));    %
Determining the optimal configuration
        [obj.PcuSec, indexSec] =
min(PsecAC(:,indexInter(indexAw),indexAw));
        [obj.PcuPri, indexPri] =
min(PpriAC(:,indexInter(indexAw),indexAw));
        obj.Ninterleave = indexInter(indexAw);
        obj.AwPri = AwPriM(indexAw);
        obj.AwSec = AwSecM(indexAw);
        obj.wPri = litzWires(indexPri);
        obj.wSec = litzWires(indexSec);
        obj.JPri = obj.Ipri/obj.AwPri;
        obj.JSec = obj.Isec/obj.AwSec;
        obj.Ptot = obj.Pcore + obj.Pcu;
        obj.eta = (obj.Po-obj.Ptot) / obj.Po*100;

    end
end
end

```

```

classdef LitzWire < Wire
    %LITZWIRE Class that holds information of the Litz Wire to be
    used
    % Detailed explanation goes here

    properties
        Ao      % Total Area (not copper area)
        Aw      % Total wire Area
        Fr      % Rac/Rdc (will have two values(one for primary, one
for secondary)
        Nstrand % Number of strands in the Litz Wire
        OD      % Outer Diameter of wire + 10% safety factor
        Rac     % AC Resistance
        Rdc     % DC Resistance of bundle
    end

    methods
        function obj = LitzWire(objArray, Ns,b, sd, len)
            %LITZWIRE Construct an instance of this class
            % Detailed explanation goes here
            if nargin ~= 0
                m = size(objArray,1);
                obj(m) = obj; % Create an array of objects
                for i=1:m
                    obj(i).Nstrand = objArray(i,1);
                    obj(i).awg = objArray(i,2);
                    obj(i).d = objArray(i,3);
                    obj(i).a = objArray(i,4);
                    obj(i).len = len;
                    obj(i).R = obj(i).calcR();
                    obj(i).Fr = obj(i).calcFr(Ns, b, sd);
                    obj(i).Rdc = obj(i).calcRdc();
                    obj(i).Rac = obj(i).calcRac();
                    obj(i).OD = 1.1*(1.28 *
sqrt(obj(i).Nstrand)*obj(i).d);
                    obj(i).Ao = pi*(obj(i).OD/2)^2;
                    obj(i).Aw = obj(i).Nstrand * obj(i).a;

```

```

        end
    end
end
function Fr = calcFr(obj,Ns,b, sd)
    Fr = 1+((pi.*obj.Nstrand.*Ns).^2.*(obj.d).^6)...
        ./(192*(sd.*1000).^4.*b^2);
end
function Rdc = calcRdc(obj)
    Rdc = obj.R./obj.Nstrand;
end
function Rac = calcRac(obj)
    Rac = obj.Rdc .* obj.Fr;
end
function Rdc = getRdc(obj, index)
    Rdc = zeros(1,length(obj));
    for i = 1:length(obj)
        Rdc(i) = obj(i).Rdc(index);
    end
end
function Rac = getRac(obj,index)
    Rac = zeros(1,length(obj));
    for i = 1:length(obj)
        Rac(i) = obj(i).Rac(index);
    end
end
end
end

```

Wire.m

```

classdef Wire
    %WIRE Summarizes property of Wire
    % Detailed explanation goes here

    properties
        a          % Area of wire (mm^2)
        awg        % Wire Gauge number
    end
end

```

```

        d      % Diameter of wire (mm)
        len    % Required Length of wire for T
        R      % Single Wire Resistance
        rho_cu = 1.68e-8; % Copper Resistivity (ohm m)

    end

    methods
        function obj = Wire(awg, d, a)
            %WIRE Construct an instance of this class
            % Detailed explanation goes here
            if nargin ~= 0
                obj.awg = awg;
                obj.d = d;
                obj.a = a;
            end

        end

        function R = calcR(obj)
            R = (obj.rho_cu .* obj.len ./ (obj.a*0.001^2));
        end

    end
end
end

```

getSkinDepth.m

```

function [skinDepth] = getSkinDepth(f)
%GETSKINDEPTH Returns the skip depth, assuming the material is made
out of
%copper
% Detailed explanation goes here
rho = 1.678e-8; % resistivity of copper (ohm *m)
mur = 0.999991; % relative permeability
mu0 = 4*pi*1e-7; % free space permeability
skinDepth = sqrt(rho / (pi*f*mur*mu0));
end

```

Core.m

```
classdef Core
    %CORE Summary of this class goes here
    % Detailed explanation goes here

    properties
        CF      % Core Factor (mm^-1)
        Le      % Effective magnetic path length (mm)
        Ae      % Effective Cross-Sectional Area (mm^2)
        Ve      % Effective Core Volume (mm^3)
        Aw      % Winding Area
        Ww      % Minimum winding width
        LN      % Average Turn Length (mm)
        m       % Mass (g/set)
        AP      % Area Product
        mui     % Initial Permeability
        Bsat    % Saturation Magnetic Flux Density (Estimate)
        Al      % AL-Value @ 1kHz, 0.5mA(nH/N^2)
        AlOpt   % Optimal Al
        CL      % Core Loss @ 80C, 300kHz, W/g
        g       % Air gap in
        gOpt    % Optimal Air Gap
    end

    methods
        function obj = Core(CF, Le, Ae, Ve, Aw, Ww, LN, m, mui, ...
                           Bsat, Al, CL, g)
            %CORE Construct an instance of this class
            % Detailed explanation goes here
            obj.CF = CF;
            obj.Le = Le;
            obj.Ae = Ae;
            obj.Ve = Ve;
            obj.Aw = Aw;
            obj.Ww = Ww;
            obj.LN = LN;
```

```

        obj.m = m;
        obj.AP = getAP(Ae,Aw);
        obj.mui= mui;
        obj.Bsat = Bsat;
        obj.A1 = A1;
        obj.CL = CL;
        obj.g = g;

    end

    function Pcore = getCL(obj,Bact)
        %METHOD1 Summary of this method goes here
        % Detailed explanation goes here
        Pcore = obj.m * interp1(obj.CL(1,:)','obj.CL(2,:)',' Bact);
    end
end
end

```

getAP.m

```

function [AP] = getAP(Ae,Aw)
%getAP Returns Area Product in cm^4 from mm^2 inputs
% Detailed explanation goes here
AP = (Ae*.01*Aw*.01);
end

```

getCL.m

```

function CL = getCL(CL_table,B)
%GETCL Returns the core loss for a given B using interpolation
% Detailed explanation goes here
CL = interp1(CL_table(1,:)','CL_table(2,:)',' B);
end

```

Timeline of Tasks and Milestones

Table 8-1: Final Board Bill of Materials

Manufacturer Part Number	Manufacturer	Customer Reference	Reference Designator	Quantity	Unit Price	Extended Price	Total Price
FDP083N15A-F102	ON Semiconductor	Catch Mosfet x2	M3	1	4.32	\$4.32	\$66.79
ACPL-W346-500E	Broadcom Limited	2.5A Gate Driver Optical Coupling 5000Vrms 1 Cha	U2	1	5.05	\$5.05	
C4532X7T2J304K2 50KA	TDK Corporation	Capacitor MLCC 0.3uf 630V CAP 1812/4532	C27, C28, C29, C30	4	1.252	\$5.01	
IHLP6767GZER1R5 M01	Vishay Dale	1.5µH Shielded Molded Inductor 42A 1.88mOhm Max	L2	1	5.87	\$5.87	
EEH-ZA1E221P	Panasonic Electronic Components	Capacitor Aluminum Hybrid Radial 330uf 25V CAP E	C5	1	2.7	\$2.70	
7.44E+08	Wurth Electronics Inc.	FIXED IND 4.7UH 6A 19.5 MOHM SMD	L1	1	3.6	\$3.60	
IPP60R190P6XKSA 1	Infineon Technologies	Clamp Transistor	M5	1	3.23	\$3.23	
CPC1393GRTR	IXYS Integrated Circuits Division	Solid State Relay SPST-NO (1 Form A) 4- SMD (0.30	U6	1	2.63	\$2.63	
CL10B474KA8NFN C	Samsung Electro- Mechanics	Capacitor MLCC 0.47uf 25V CAP 0603/1608	C18	1	0.073	\$0.07	
PS2802-1-A	Renesas Electronics America	PC817B	U5	1	1.52	\$1.52	
C3216X7R1E106K1 60AB	TDK Corporation	Capacitor MLCC 10uf 25V CAP 1206/3216	C6, C7, C8, C9	4	0.566	\$2.26	
B66359B1013T001	EPCOS (TDK)	BOBBIN COIL FORMER ETD 29X16X10	T2	1	1.84	\$1.84	

TPHR6503PL,L1Q	Toshiba Semiconductor and Storage	Forward Mosfet	M2	1	1.82	\$1.82	
IPN80R750P7ATM A1	Infineon Technologies	AUX Transistor	M1	1	1.64	\$1.64	
EKXJ501ELL100MJ 30S	United Chemi-Con	10µF 500V Aluminum Electrolytic Capacitors Radia	C26	1	1.03	\$1.03	
HRG3216P-5600-D- T1	Susumu	RES 560 OHM 1% 1W 1206	R2	1	0.97	\$0.97	
TLR2H15DR001FT DG	TE Connectivity Passive Product	RES 0.001 OHM 1% 1.5W 2010	R3	1	0.97	\$0.97	
B66359S2000X000	EPCOS (TDK)	Supplier Device Package ETD 29 x 16 x 10	T2	1	0.59	\$0.59	
PT6WV-471A2020	Amphenol Piher Sensing Systems	TRIMMER 470 OHM LINEAR	R36	1	0.81	\$0.81	
251R15S110GV4E	Johanson Technology Inc.	Capacitor MLCC 11pf 250V CAP 0805/2012	C3	1	0.73	\$0.73	
PMR18EZPFU8L00	Rohm Semiconductor	RES 0.008 OHM 1% 1W 1206	R5	1	0.73	\$0.73	
SD560BTR	SMC Diode Solutions	DIODE GEN PURP 600V 5A SMB	D7	1	0.51	\$0.51	
C1210C473KBRAC 7800	KEMET	Capacitor MLCC 50.047uf 630V CAP 1210/3225	C14	1	0.49	\$0.49	
BAT46WJ,115	Nexperia USA Inc.	BAT46WJ	D1, D2	2	0.39	\$0.78	
TMK316BBJ226ML- T	Taiyo Yuden	Capacitor MLCC 22uf 25V CAP 1206/3216	C2	1	0.58	\$0.58	
B66358G0200X187	EPCOS (TDK)	Uncoated N87 Ferrite Core ETD Type 30.60mm Lengt	T2	2	0.99	\$1.98	

GRM21BR61E226 ME44L	Murata Electronics North America	Capacitor MLCC 22uf 25V CAP 0805/2012	C16	1	0.49	\$0.49	
CL10B224KB8VPN C	Samsung Electro- Mechanics	Capacitor MLCC 0.22uf 50V CAP 0603/1608	C11, C15	2	0.21	\$0.42	
LH R974-LP-1	OSRAM Opto Semiconductors Inc.	LED RED DIFFUSED 0805 SMD	D5	1	0.33	\$0.33	
CL21A475KAQNNN E	Samsung Electro- Mechanics	Capacitor MLCC 4.7uf 25V CAP 0805/2012	C1, C17, C21	3	0.129	\$0.39	
CL10X106MO8NRN C	Samsung Electro- Mechanics	Capacitor MLCC 10uf 16V CAP 0603/1608	C10	1	0.29	\$0.29	
SR0805JR-7W10RL	Yageo	Resistor 10 0805 1/4W 10%	R16	1	0.27	\$0.27	
1N4148W-7-F	Diodes Incorporated	DIODE GEN PURP 100V 300MA SOD123	D4, D6	2	0.16	\$0.32	
RS1MTR	SMC Diode Solutions	DIODE GEN PURP 1KV 1A SMA	D3	1	0.17	\$0.17	
RNCP0805FTD1K0 0	Stackpole Electronics Inc	Resistor 1.0k 0805 1/4W 5%	R25	1	0.072	\$0.07	
CC0805JRNPOAB N680	Yageo	Capacitor MLCC 68pf 200V CAP 0805/2012	C4	1	0.18	\$0.18	
CC0603JRNPO8BN 221	Yageo	Capacitor MLCC 220pf 25V CAP 0603/1608	C25	1	0.069	\$0.07	
TMK107B7223KA-T	Taiyo Yuden	Capacitor MLCC 0.022uf 16V CAP 0603/1608	C22	1	0.068	\$0.07	
ERJ-3EKF4990V	Panasonic Electronic Components	Resistor 499 0603 1/10W 1%	R14	1	0.066	\$0.07	
CL10B105KA8NNN C	Samsung Electro- Mechanics	Capacitor MLCC 1.0uf 25V CAP 0603/1608	C24	1	0.06	\$0.06	

MCR18ERTFL1R30	Rohm Semiconductor	RES 1.3 OHM 1% 1/4W 1206	R6	1	0.1	\$0.10	
RNCP0805FTD4R70	Stackpole Electronics Inc	Resistor 4.7 0805 1/4W 1%	R7	1	0.1	\$0.10	
CRM0805-FX-51R0ELF	Bourns Inc.	Resistor 51 0805 1/4W 1%	R41	1	0.1	\$0.10	
RMCF1206FT715K	Stackpole Electronics Inc	Resistor 715k 1206 1/4W 1%	R15	1	0.036	\$0.04	
CRGCQ0603F150R	TE Connectivity Passive Product	Resistor 150 0603 1/10W 5%	R10	1	0.031	\$0.03	
CRGCQ0603F560R	TE Connectivity Passive Product	Resistor 560 0603 1/10W 1%	R40, R43	2	0.031	\$0.06	
CL10B104KA8NNNC	Samsung Electro-Mechanics	Capacitor MLCC 0.10uf 25V CAP 0603/1608	C19	1	0.029	\$0.03	
8.85E+11	Würth Electronics Inc.	Capacitor MLCC 68pf 25V CAP 0603/1608	C20	1	0.027	\$0.03	
8.85E+11	Würth Electronics Inc.	Capacitor MLCC 0.033uf 16V CAP 0603/1608	C23	1	0.026	\$0.03	
8.85E+11	Würth Electronics Inc.	Capacitor MLCC 470pf 50V CAP 0603/1608	C13	1	0.023	\$0.02	
CRGCQ0603J220R	TE Connectivity Passive Product	Resistor 220 0603 1/10W 5%	R9	1	0.022	\$0.02	
RMCF0603FT360R	Stackpole Electronics Inc	Resistor 360 0603 1/10W 1%	R32, R33, R37	3	0.017	\$0.05	
RMCF0603FT806R	Stackpole Electronics Inc	Resistor 806 0603 1/10W 1%	R12	1	0.017	\$0.02	
RMCF0603FT820R	Stackpole Electronics Inc	Resistor 820 0603 1/10W 1%	R35	1	0.017	\$0.02	

RMCF0603FG1M00	Stackpole Electronics Inc	Resistor 1.0M 0603 1/10W 1%	R17	1	0.017	\$0.02	
RMCF0603FT1K50	Stackpole Electronics Inc	Resistor 1.5k 0603 1/10W 1%	R20, R42	2	0.017	\$0.03	
RMCF0603FT100K	Stackpole Electronics Inc	Resistor 100k 0603 1/10W 1%	R22, R23	2	0.017	\$0.03	
RMCF0603FT10K0	Stackpole Electronics Inc	Resistor 10k 0603 1/10W 1%	R11, R34, R38, R8	4	0.017	\$0.07	
RMCF0603FT15K0	Stackpole Electronics Inc	Resistor 15k 0603 1/10W 1%	R30	1	0.017	\$0.02	
RMCF0603FT2K80	Stackpole Electronics Inc	Resistor 2.8k 0603 1/10W 1%	R13	1	0.017	\$0.02	
RMCF0603FT20K0	Stackpole Electronics Inc	Resistor 20k 0603 1/10W 1%	R18, R19	2	0.017	\$0.03	
RMCF0603FT24K9	Stackpole Electronics Inc	Resistor 24.9k 0603 1/10W 1%	R26	1	0.017	\$0.02	
RMCF0603FT3K00	Stackpole Electronics Inc	Resistor 3.0k 0603 1/10W 1%	R24	1	0.017	\$0.02	
RMCF0603FT36K5	Stackpole Electronics Inc	Resistor 36.5k 0603 1/10W 1%	R21	1	0.017	\$0.02	
RMCF0603FT39K0	Stackpole Electronics Inc	Resistor 39k 0603 1/10W 1%	R4	1	0.017	\$0.02	
RMCF0603FT47K0	Stackpole Electronics Inc	Resistor 47k 0603 1/10W 1%	R39	1	0.017	\$0.02	
RMCF0603FT51K0	Stackpole Electronics Inc	Resistor 51k 0603 1/10W 1%	R28	1	0.017	\$0.02	
RMCF0603FT68K0	Stackpole Electronics Inc	Resistor 68k 0603 1/10W 1%	R27	1	0.017	\$0.02	

RMCF0603FT7K50	Stackpole Electronics Inc	Resistor 7.5k 0603 1/10W 1%	R31	1	0.017	\$0.02	
RMCF0603FT76K8	Stackpole Electronics Inc	Resistor 76.8k 0603 1/10 1%	R29	1	0.017	\$0.02	
GRM188R61E225K A12D	Murata Electronics North America	Capacitor MLCC 2.2uf 25V CAP 0603/1608	C12	1	0.17	\$0.17	
V6560X	Assmann WSW Components	1	na	1	0.67	\$0.67	
578622B03200G	Aavid, Thermal Division of Boyd Corporation	1	na	1	1.4	\$1.40	
NMS-306	Essentra Components	1	na	3	0.171	\$0.51	
HN-M3-79	Essentra Components	1	na	3	0.167	\$0.50	
CSRF0603FT2L50	Stackpole Electronics Inc	2.5 mOhms $\pm 1\%$ 0.25W, 1/4W Chip Resistor 0603 (16	R1	1	0.63	\$0.63	
XG8T-0231	Omron Electronics Inc-EMC Div		W1,W2	2	0.19	\$0.38	
QPC02SXGN-RC	Sullins Connector Solutions		W1,W2	2	0.1	\$0.20	
DDZ9701-7	Diodes Incorporated	Zener Diode 14V 500mW $\pm 5\%$ Surface Mount SOD-123	D8	1	Sample	\$0.00	
04853.15DR	Littelfuse	FUSE BOARD MNT 3.15A 600VDC 2SMD	F1	1	Sample	\$0.00	
74404020100	Würth Electronics Inc.	10 μ H Shielded Wirewound Inductor 500mA 850mOhm Nonstandard	L3	1	Sample	\$0.00	
750817020	Würth Electronics Inc.	Auxilliary Transformer for DC-DC Senior Project	T1	1	Sample	\$0.00	

PE-68386NLT	Pulse Electronics Power	PULSE XFMR 1:1 785UH	T3	1	Sample	\$0.00	
5029	Keystone Electronics	Test point	TP1 Thru TP35	32	Sample	\$0.00	
XLMG3410R050R WHT	Texas Instruments	N-Channel 600-V 50mOhm GaN with integrated driver and protection	U1	1	Sample	\$0.00	
LT3752FE-1	Linear Technology	Forward Converter Regulator Positive, Isolation Capable Output Step-Up/Step-Down DC-DC Controller IC 6.5-100V Input	U3	1	Sample	\$0.00	
LT8311FE	Linear Technology	Power Supply Controller Secondary-Side Controller	U4	1	Sample	\$0.00	
43650-0402	Molex		X1	1	Sample	\$0.00	
7466113	Würth Electronics Inc.	1 Pin Shank Terminal M3 Surface Mount	X2, X3	2	Sample	\$0.00	
N/A	Mike's Electronics Parts	6604/44 Litz Wire	T2	1.558656	1.294	\$2.02	
	Mike's Electronics Parts	420/46 Litz Wire	T2	5.715072	0.764	\$4.37	

Analysis of Senior Project Design

a) Summary of Functional Requirements

This project will accept a 200-420VDC input from the Cal Poly Racing, Formula SAE Electric Vehicle's battery pack and output 12-15VDC at loads up to 500W.

The output voltage will be programmable and telemetry data reporting the status of the DC-DC's performance will be outputted. Many shutdown conditions are possible for the DC-DC. Overcurrent, overvoltage, and reverse voltage protection circuits will be included on the input and output in order to shut the device down. Additionally, a discrete shutdown/enable input will be included in order to disable the device when necessary.

b) Primary Constraints

The primary constraint for the design of the DC-DC is the Formula SAE Rulebook. Nearly all safety requirements for the DC-DC are derived the Formula SAE Rulebook. Failure to abide by these constraints will result in a failure to be useful for our largest market. Furthermore, failure to follow these rules will also significantly degrade the general safety of the product. Additionally, the system must be able to operate under harsh environmental conditions, such as rain and high temperature. Finally, the DC-DC must be small and light in order to be competitive in the automotive industry.

c) Economic

What economic impacts result?

i) Human Capital

This device will create new jobs in engineering, manufacturing, sales and technical support.

ii) Financial Capital

This product will not create a lot of business for suppliers because small-series EV manufacturers and Formula SAE teams represent a very small portion of the total EV market. However, this product will assist the customer due to its \$40-\$200 price improvement.

iii) Natural Capital

Aluminum sheet or plastic will be required to create the enclosure of the DC-DC. FR-4 material and a board fab house will be needed to create the main board. Finally, the stock from various IC and passive companies are needed to populate the board. Significant amounts of water is used in to create ICs.

When and where do costs and benefits accrue throughout the project lifecycle?

Costs to the senior project team includes, design costs, manufacturing costs, testing costs, and material costs. Costs to the customer include the upfront cost, repackaging costs, harness integration cost, and the code costs. These will all occur at the initial implementation of the product. After successful integration, the customer will benefit from improved efficiency, reduced weight, and increased packaging headroom throughout the lifetime of the product.

What inputs does the experiment require? How much does the project cost? Who Pays?

Table 8-2: Original Cost Estimate

Item	Quantity	Cost
DAW System/Switching Controllers	1	\$15
Switching Device	8	\$5
Gate Drivers	1	\$50
Magnetics	1	\$50
Other ICs		\$60
Printed Circuit Board	1	\$150
Connectors, Enclosure, and Hardware		\$60

With a very aggressive budgeting, we believe that a total cost of approximately \$400 will be required to develop this project. These prices are within the \$200 per person grant given to EE senior projects. Moreover, we expect to be able to nullify a large portion of the costs through existing materials in the Cal Poly Racing inventory, and partnerships with Linear Technology and Bay Area Circuits.

How much does the product earn? Who profits?

The predicted cost of manufacturing is approximately \$95. To obtain a margin of 40%, we will be selling our device at \$130. Therefore, each device will result in \$35 of profit. The profit from selling of these devices will be split between the two designers of the project: Nick Mah and Jason Zhou. We expect

to sell 20-30 units to other Formula SAE teams, leading to a potential profit of \$800-\$1200.

Timing

The DC-DC will be released after the senior project expo. Afterwards, we will begin marketing to Formula SAE teams, while probing the commercial market for interest in our device. If there is interest, we will begin further testing to certify our device for commercial use. After completion of testing, the device will immediately be released to Cal Poly Racing, as they are the primary contractor.

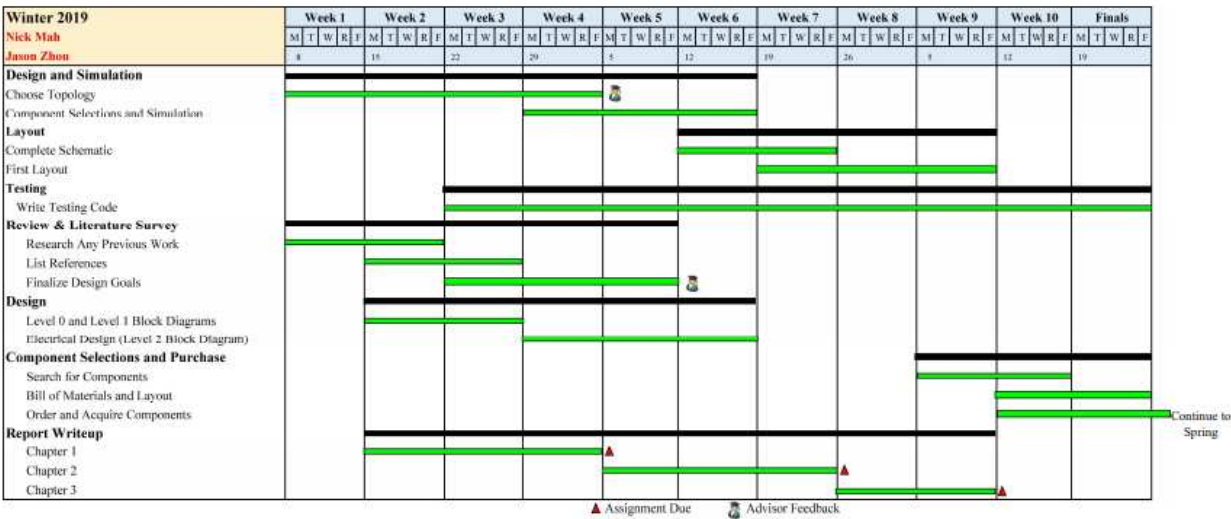


Figure 8-3: Winter Quarter Timeline

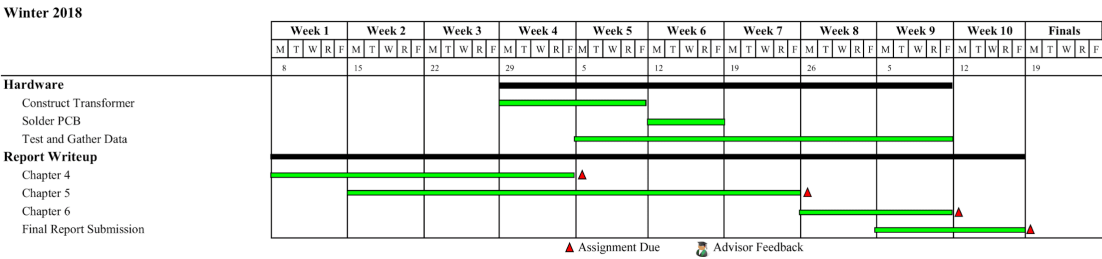


Figure 8-4: Spring Quarter Timeline

d) If Manufactured on a Commercial Basis:

Estimated number of devices sold per year: 500 (40-50 from Formula SAE and Formula Student, the rest from small EV manufacturers)

Estimated Manufacturing Cost for Each Device: \$95

Estimated Purchase Price for Each Device: \$130

Estimated Profit Per Year: \$17500

*Estimated Cost for User to Operate Device: $\frac{Cost}{Hour} = C_{elec} * P_{max} * \eta = \frac{\$0.12}{kWh} *$*

$$0.5kW * 0.9 = \frac{\$0.54}{hour}$$

e) Environmental

This product will impact the environment at all points of its use. The creation of the DC-DC will involve the manufacturing of electrical components, PCBs and metal enclosures. All of the above stress the environment with water usage, waste chemicals and materials, emissions and energy usage. During usage, the efficiency of the device will cause some wasted electricity. Since this is an electric vehicle-oriented device, the environmental impacts of electric vehicles should also be considered. The large battery packs of electric vehicles can lead to very large electrical waste deposits. On the other hand, the high efficiency of electric vehicles could offset the pollution costs associated with ICE vehicles. Finally, at end of life, the device will need to be recycled. Because the device is used in student-built vehicles. There is a high chance of proper electronics recycling occurring. Additionally, because of the big ewaste hazard electric

vehicles pose, there is a strong likelihood that the DC-DC will be properly recycled along with the rest of the EV.

f) Manufacturability

A lot of effort will need to be put into the construction of the device.

Although vibration and temperature environments are not as extreme as in an ICE vehicle, a high amount of care must be put into the reliability of the device.

Therefore, our manufacturing process will require rigorous vibration, thermal, endurance, and isolation testing throughout the manufacturing process to ensure customer safety.

g) Sustainability

Describe any issues or challenges associated with maintaining the completed device or system

The primary challenge with maintaining the device is environmental conditions. Automobiles undergo very harsh environmental and startup conditions. Our DC-DC must be able to withstand high vibration and ambient temperatures as well as constant power cycling. Additionally, a balance between an environmentally proof enclosure and effective thermal dissipation will need to be found.

Describe how the project impacts the sustainable use of resources

This product stresses the natural resources during production and end of life. The primary stresses are water usage, ewaste, pollution/chemical emissions, and energy usage. However, successful implementation of the DC-DC could

result in increased EV usage, which would help reduce the total pollution impact of personal transportation.

Describe any upgrades that would improve the design of the project

In order to further reduce size, water cooling could be included to decrease the required cooling effort. This would allow designers to use smaller switching IC's, which should improve efficiency, as the switching energy will decrease. This would also decrease the total required mass for heatsinks.

Another potential improvement is the addition of a complete microcontroller system. This would significantly improve the tunability of the design as well as the amount of telemetry that could be sent back.

A final potential improvement is the splitting the DC-DC into two, separate, lower power outputs. By having two separate outputs, more flexibility of design is afforded. The output has two separate output voltages, reducing the amount of voltage regulators required on other boards in the car.

Describe any issues or challenges associated with upgrading the design

Water-cooling will increase the difficulty of manufacturing significantly. Another issue with water-cooling is that we lose flexibility from our value proposition.

The addition of a microcontroller will reduce the ease of integration for our device. Additionally, our personal costs of development time and manufacturing costs will go up now that we will need to support a more complicated firmware package.

The addition of a microcontroller will reduce the ease of integration for our device. Additionally, our personal costs of development time and manufacturing costs will go up now that we will need to support a more complicated firmware package.

Splitting of the outputs will improve the flexibility of design. However, it will increase the size and weight of the DC-DC, which is a primary component of our value proposition.

h) Ethical

The DC-DC has multiple ethical concerns. The DC-DC must be made such that it is compliant with the Formula SAE Rulebook. Workarounds that "technically" pass the rules are also unethical as they betray the intent of the rule. Additionally, the DC-DC must be extremely reliable. Any product going into automotive use is purchased in faith that the device will work reliably. Any failure modes possible with the DC-DC must be documented for the customer. Any failure modes that could result in potentially dangerous situations must be accounted for and prevented before release to the customer. An example of a dangerous failure mode is a loss of isolation, which could result in the death of the driver. Unfortunately, the designers of the product cannot control how the DC-DC is operated. Modifications, such as reflashing the controller, could result in a harmful system. Finally, usage of the device must be safe. High voltages are present at the input of the device. The designers must be cognizant of this danger and make installation safe and well documented.

i) Health and Safety

The primary safety concern for the DC-DC are failure modes and interaction with the DC-DC. Loss of isolation of the DC-DC could result in permanently damaging electric shock. Additionally, if the failure of a component is not well contained by the enclosure, it could result in the destruction of nearby electronics. Finally, connecting the input leads to the DC-DC could result in electric shock as well if proper installation procedures are not followed.

j) Social and Political

Because this device is designed to be completely Formula SAE rules compliant while maintaining a competitive price compared to existing solutions, the competition may see a decrease in revenue. This can affect the livelihood of employees. Because integration of the device is very simple, we expect our DC-DC to help equalize the disparity between well-funded and underfunded race teams.

k) Development

This project will require development in knowledge of thermal dissipation in electronics, isolated power supply design, power electronics, environmental-rated components, automated testing, and magnetic design.